

When credibility becomes the policy

Three rigorous economics papers · two current applied-development experiments and one land-rights classic. The following article pages are edited reading excerpts, not complete replacements for the originals.

LITERATURE PULSE

These papers ask when an official signal becomes believable enough to change behavior. In Tunisia, information moved organ-donor registration when it also built trust in the medical institution delivering it. In Colombia, a government skill certificate improved employment for local workers but not detectably for Venezuelan migrants, whose credentials and labor-market position start from a different baseline. Rwanda's land-title reform made investment and women's claims more legible, but only legally married women gained. **Literature trend:** applied development economics is increasingly treating credibility as an economic input rather than background context. Random assignment and a spatial discontinuity identify policy effects here, while the heterogeneous results show why legal status, family approval, and institutional trust belong inside the mechanism—not just in the limitations section.

01 CURRENT / PUBLIC HEALTH AND TRUST · STARTS ON PAGE 2 · Journal of Development Economics · online first 2026 · peer reviewed · CC BY 4.0

Fostering Institutional Trust and Organ Donation: Evidence from a Randomized Informational Intervention in Tunisia

Can reliable information and a trusted messenger turn stated support into an actual health decision? In 57 Tunisian university classrooms, the authors randomized a transplant-survivor video, a doctor's presentation, and a Q&A.; The intervention raised deceased-donor registration by 2.7 percentage points, medical and legal knowledge by more than 0.8 standard deviations, and institutional trust by about 0.5 standard deviations. Effects were concentrated among men and students expecting family approval. The university sample, family veto power, and attrition at six months limit generalization, but the study clearly shows that trust can be a productive input into public-health take-up.

02 CURRENT / MIGRATION AND LABOR · STARTS ON PAGE 19 · Journal of Development Economics · accepted 2026 · peer reviewed · CC BY 4.0

Skill Certifications and Migrant Labor Market Integration: Experimental Evidence from Colombia

Can a government credential make workers' existing skills legible to employers? Among 2,160 eligible applicants in Colombia, 855 were randomly offered occupation-specific assessment and certification. The offer raised local workers' employment by 5.7 percentage points and weekly hours by about two, but did not raise wages or formal employment. For Venezuelan migrants, estimates were imprecise and showed no detectable gains; the study cannot rule out smaller effects. Differential attrition and bundled participation subsidies are important caveats. The striking result is that the same signal worked differently for groups with very different baseline credibility in the labor market.

03 CLASSIC / LAND RIGHTS AND GENDER · STARTS ON PAGE 35 · Journal of Development Economics 110 · 2014 · peer reviewed classic · CC BY 3.0 IGO edition

Environmental and Gender Impacts of Land Tenure Regularization in Africa: Pilot Evidence from Rwanda

Can low-cost land registration improve investment without dispossessing vulnerable owners? The classic studies Rwanda's pilot program with a geographic discontinuity and spatial fixed effects, comparing roughly 3,500 households near cell boundaries. Registration nearly doubled soil-conservation investment, raised documented joint ownership for legally married women by 17 percentage points, and clarified inheritance without triggering distress sales. But unmarried women lost documented claims, and the cross-sectional boundary design identifies a local, short-run effect. The paper remains a model of applied work that treats efficiency and distribution as outcomes of the same institutional reform.

Editorial note: references and appendices were omitted where possible; no included article contributes more than 15 text pages. Full citations and source links remain in the catalog. AI-written synopses were checked against the included source pages.

Fostering Institutional Trust and Organ Donation: Evidence from a Randomized Informational Intervention in Tunisia*

Christina Sarah Hauser[†], Maleke Fourati[‡]

August 21, 2026

Abstract

Organ transplantation saves lives and improves the quality of life of patients who would otherwise depend on costly and invasive treatments. The technology is becoming available in low- and middle-income countries, but the gap between the demand for and the supply of organ transplants remains large. In a randomized controlled trial at a Tunisian university, we evaluate an expert-led intervention designed to inform about organ donation and strengthen trust in medical institutions. The intervention consists of a short video testimonial from a heart transplant survivor, a presentation led by a medical doctor and a short Q&A session. We find that the intervention significantly increases deceased donor registration, with suggestive evidence of positive spillover effects on the control group. The survey data provides little support for changes in attitudes and social expectations as potential mechanisms. Instead, it reveals large and statistically significant increases in medical and legal knowledge, religious beliefs supportive of donation, and institutional trust among treated students. Finally, we show that treatment effects on organ donor registration are driven by male students and those whose families likely support organ donation. Targeting older segments of the population may further increase the impact and cost-effectiveness of this intervention.

JEL classification: I12, I15, O17, F63

Keywords: organ donation; health; institutional trust; peer effects

*This study was funded by the Weiss Fund for Research in Development Economics at the University of Chicago [Implementation & Policy Grant, July 2024 and May 2025] and by the European University Institute. The study received IRB approval by the Ethics Committee of the European University Institute on 3 April 2024 (20240227-HAUSER). The study was pre-registered at the [AEA Social Science registry](#) on 7 March 2025 (AEARCTR-0015516) and was conditionally accepted [via pre-results review](#) at the *Journal of Development Economics* in March 2025.

[†][Corresponding author]. European University Institute and Collegio Carlo Alberto. Permanent address: University College Dublin, School of Economics, Belfield, Dublin 4, Ireland. Email: christina.hauser@ucd.ie.

[‡]South Mediterranean University, Mediterranean School of Business. 1053 Les Berges du Lac II Walkway, Tunis 1053, Tunisia. L.R. MASE (LR21ES21), ESSAI, Université de Carthage. Economic Research Forum. Email: maleke.fourati@msb.tn.

1 Introduction

Organ transplantation is a cost-effective way to treat organ failure due to genetic conditions, infections such as hepatitis, or chronic diseases such as diabetes (Olawade et al.; 2025). The infrastructure and medical expertise necessary to harvest, store and transplant organs is available in many countries, including an increasing number of low- and middle-income countries (LMICs) (GODT; 2024). Due to ethical and legal concerns, there is no regular competitive market for organs.¹ Unsurprisingly, the demand for donated organs tends to exceed the supply by far. Patients with organ failure face a significantly reduced life expectancy (Stokes; 2011), and if alternative treatments, such as hemodialysis, are possible, they tend to be intrusive and painful for the patient, and costly for health systems. In most cases, patients in need of a transplant die if the transplant is not available on time (HDSA; 2024). The number of deceased organ donors is particularly low in LMICs (GODT; 2024). Based on qualitative and quantitative evidence from Tunisia, we conjectured that a general lack of awareness, religious misperceptions, and low institutional trust contribute to this organ transplant shortage. This study asks whether an expert-led informational intervention can correct misperceptions, foster institutional trust, and ultimately increase the number of deceased organ donor registrations.

Economists have designed and evaluated different policies to increase the supply of donated organs. Nobel Prize laureate Alvin Roth and co-authors' research on kidney exchange (Roth et al., 2004; Roth et al., 2005) and non-directed deceased donor chains (Rees et al., 2009; Anderson et al., 2015) has contributed to the establishment of a formal kidney exchange in the United States. Another policy that has been investigated in the laboratory (Kessler and Roth, 2012; Herr and Normann, 2016) and successfully implemented in Israel, is giving priority to registered donors and their families (Stoler et al.; 2016). Other studies have focused on price mechanisms and tested under which conditions organ markets may be socially acceptable in the United States (Becker and Elías, 2007; Elías et al., 2015; Elías et al., 2019; Becker et al., 2022). In continental Europe, "presumed consent" has proven to be successful in increasing the supply of donated organs (Johnson and Goldstein, 2003; Abadie and Gay, 2006; Steffel et al., 2019). Presumed consent makes being a donor the default option: an individual is considered a potential donor *unless* they have actively opted

¹The sale of organs is illegal in most countries, with the notable exception of Iran, which allows for kidney sales among its citizens (Moghaddasi and Sgroi; 2024). All other health systems rely on organ donation and, to a lesser degree, on kidney exchange (Roth et al.; 2004).

out of the system by declaring themselves a non-donor. However, the success of this policy relies on high levels of institutional trust (Shepherd et al.; 2014). It is thus unlikely to be a solution in many LMICs, where trust in institutions tends to be low (e.g. Algan and Cahuc; 2010). Similarly, granting priority to registered donors and setting up kidney exchanges require substantial institutional capacity, and will therefore be viable policy options only in the long run.

Certain constraints to organ donation are likely to be more pronounced in LMICs. Our quantitative and qualitative research in Tunisia revealed a general lack of awareness, institutional mistrust and widespread fear of organ trafficking. In a national survey, Aissi et al. (2024) find that 73% of Tunisian adults believe that there is organ trafficking in Tunisia. Some people expressed the fear that organ donation may conflict with their religion. In reality, religious scholars across the Muslim world agree that Islam encourages organ donation, with reproductive organs being the only exception.² These context-specific constraints and misperceptions have not been sufficiently addressed by the previous literature. A lack of research on which policies work in non-Western countries may lead to the adoption of inefficient policies, harming patients and societies at large. At the same time, providing new information and correcting misperceptions can be a very cost-efficient intervention in many contexts — without requiring institutional change (Jensen, 2010; Dupas, 2011; Wiswall and Zafar, 2014; Dupas et al., 2018; Bursztyn et al., 2020; Banerjee et al., 2024a).

To address the general lack of awareness about organ transplantation and fear of organ trafficking among Tunisian adults, we designed an informational intervention in partnership with the Tunisian National Center for the Promotion of Organ Transplantation (CNPTO). We first showed a short video testimonial of a patient from a village in north-western Tunisia who had survived thanks to a heart transplant. The main intervention which followed was led by experienced medical doctors and technicians of the CNPTO. They explained the social significance of organ donation, the medical procedure and the legal and administrative framework in detail and also devoted several minutes to the Islamic perspective on organ donation. After the intervention, the team stayed for a short Q&A session with the partici-

²A Tunisian fatwa explicitly endorsed organ donation conditional on the consent of the donor or their family members back in 2006. An equivalent fatwa by the Al-Azhar University in Cairo, which is widely recognized within the Sunni community, followed in 2009 (Ali et al.; 2020). The prevailing consensus among Muslim religious scholars is that organ donation and transplantation for the purpose of saving lives is not only permitted but also to be encouraged. Ali et al. (2020) hypothesize that a lack of knowledge about these religious rulings and misunderstanding thereof contribute to the widespread skepticism about organ donation among Muslims, which is documented in many qualitative and quantitative studies (Khawari et al., 2005; Sharif et al., 2011; Gauher et al., 2013; Ralph et al., 2016).

pants. Within a randomized controlled trial (RCT) at a Tunisian university enrolling 1,277 students, we assessed the impact of this informational intervention on young adults' donor status. In Tunisia, adults can express their consent to be a deceased organ donor by adding the word "donor" to their ID. We measured donor status by offering on-the-spot ID changes on campus in collaboration with the CNPTO and the Tunisian Technical and Scientific Police during a designated "Organ Donation Awareness Week" on campus. All students had access to these on-the-spot ID changes, and control group students also received light-touch information about organ donation within the awareness week. This decreased the logistical costs of ID changes among the control group, which is why we view the estimated treatment effects as a lower bound. To assess spillover effects to non-treated students, we collected extensive network data at baseline. We also measured the immediate impact of the intervention on students' attitudes, social expectations, medical and legal knowledge, religious beliefs, and institutional trust with an endline survey immediately after the intervention. A follow-up survey six months later gathered further information to assess the persistence of treatment effects.

We find that the intervention substantially increased the likelihood that treated students add the donor status to their ID. Treated students are 2.7 percentage points more likely to change their ID. This corresponds to a 207% increase relative to the control group, as only 1.3% of control students changed their ID. While the percent change should be interpreted with caution, the treatment effect survives all robustness checks, including randomization inference, and remains statistically significant when correcting for multiple hypothesis testing. Importantly, the estimated treatment effect implies that the intervention is cost-effective compared to hemodialysis. The estimated cost per life-year saved amounts to less than USD 3,000, making the intervention highly cost-effective by WHO standards. This likely understates the true cost-effectiveness of the intervention for several reasons, including potential spillover effects and the fact that students in the control group benefited from some information and facilitated access to ID changes, resulting in substantially higher registration rates than would be expected based on national data.

We find no statistically significant average treatment effect on students' attitudes. By contrast, legal and medical knowledge both increase by over 0.8 standard deviations among the treated students. Religious beliefs become significantly more conducive to organ donation and institutional trust is 0.5 standard deviations higher among the treated than among the control group. This reflects higher trust in the CNPTO, an increase in the perception that organ donation is equitable, and a decrease in the perceived frequency of organ traffick-

tential donors as long as they do not register in the national refusal register. In practice, however, the CNPTO does not harvest organs without the explicit consent of the family of the deceased or the deceased themselves: Law 99-18 of March 1, 1999, regarding the national ID created the option of adding the word “donor” on the backside of one’s ID, next to the blood group.⁴ The institution in charge of changing IDs is the Tunisian Technical and Scientific police. Changing one’s ID to add the donor status requires signing the respective form and is, by law, free of charge. Falsification is very unlikely, which is the main argument in favor of the policy. However, ID changes come at a logistical cost for the individual who has to bring the relevant form and a biometric picture to a local police station, potentially queue until it is their turn, and subsequently wait for the new ID to be issued. To date, only 13,000 individuals — approximately 0.16% of the adult population — have registered as deceased organ donors by adding the status to their ID (Zannad; 2024), a figure that falls well below the registration rates seen in Europe and North America but which is broadly consistent with trends observed in other LMICs (see Figure A1 for global data on deceased organ donations).⁵

When a potential donor dies without the donor status on their ID, the CNPTO seeks consent to harvest their organs from the donor’s family. Refusal rates are high. In 2023, 58.3% of the families who were approached refused. As a result, the CNPTO harvested organs from only 19 deceased organ donors (CNPTO; 2024). Increasing the donor pool would have large marginal benefits: the 19 deceased organ donors provided organs for 52 patients in need of a transplant. In general, one deceased organ donor can save the lives of up to eight patients with organ failure and improve the lives of another 75 by tissue transplantation (HDSA; 2024). Moreover, surgeons learn by doing and become more skilled and efficient as they perform more transplantations (Magee and Pomfret; 2021). Alternative treatments, such as kidney dialysis, are intrusive and painful, as well as expensive for the national health system. For most patients, organ transplantation is more cost-effective, and higher deceased organ donation rates would therefore also decrease public spending (Axelrod et al., 2018; Zhang et al., 2023). Given the lack of donated organs, waiting lists for transplants grow: 1,506 patients are on the waiting list for a kidney transplant in Tunisia (CNPTO; 2024).

⁴Medical practice in Spain — one of the world’s leading organ donation and transplantation systems — is similar in this regard. While a presumed consent regime is formally in place, the national transplant center will always seek family consent before harvesting a potential donor’s organs (Streit et al.; 2023).

⁵Tunisia had 17 actual deceased organ donors in 2023 (GODT; 2024), which translates to 1.5 organ donors per 1 million inhabitants. As a reference point, Spain, the global champion in terms of deceased organ donation, counts 48.9 donors per 1 million inhabitants (Rada; 2024); the United States 48.0 (GODT; 2024). Germany has about 10.3 registered donors per 1 million inhabitants (Aerzteblatt; 2024).

In extreme cases, patients’ despair may lead to human rights violations: Egypt has become known as an international hub for organ trafficking from Sub-Saharan Africa and “organ transplant tourism” in the Middle East and North Africa (MENA) (COFS; 2011). Rumors about organ trafficking hurt trust in the medical system and are likely to decrease willingness to become organ donors even in neighboring countries (Abouna; 1993).

For privacy reasons, Tunisia does not have a centralized register of deceased organ donors (Zannad; 2024). Therefore, the only practical way to measure a behavioral outcome in the Tunisian context was to offer ID changes during an event in collaboration with the technical police. IDs are part of daily life and must be presented to public authorities on many occasions. We therefore interpret ID changes to add the donor status as a behavioral outcome which signals strong commitment to organ donation. The next subsection describes the details of our experimental design.

2.2 Evaluation Design

This study was implemented as a cluster randomized controlled trial at the South Mediterranean University (SMU) in Tunis. SMU is a private university and research institution, that comprises two main departments, engineering and business. In partnership with the CNPTO, we organized an “Organ Donation Awareness Week” on campus from 7–11 April 2025. The event was publicly announced and an advertisement was screened all week.⁶ This light-touch intervention was in the background of the more targeted expert-led informational intervention, which was randomized at the classroom level. We identified 42 classes without student overlap from SMU’s undergraduate programs. After gathering consent from the university and lecturers, we randomly allocated 21 classes to a treatment condition and 21 classes to a control condition, stratifying by department and year of study.

As illustrated in Panel A of Figure A2, we collected three rounds of survey data from the students. Baseline data was collected in March 2025.⁷ With a Qualtrics questionnaire, we measured students’ basic demographic characteristics, attitudes towards organ donation, altruism, risk aversion, propensity to give socially desirable answers (Reynolds; 1982), and their social network at school (similar to Alan and Kubilay; 2025; see Section D.1). We

⁶For ethical reasons, all students, lecturers and staff on campus had to receive a minimum amount of information about the event.

⁷Delays in data clearance implied that we had to run the baseline survey during Ramadan, which contributed to a relatively low response rate. The response rate, however, did not significantly differ between the treatment arms: 55.2% in the control group and 54.5% in the treatment group, p-value 0.823.

measured the immediate impact of the intervention with an endline survey in April 2025 (see Section D.2). Finally, we measured its long-term impact with a follow-up survey in October 2025 (see Section D.3).

Given low response rates to the baseline survey in March 2025, we decided to collect additional data from the incoming cohort of first year students in fall 2025.⁸ As illustrated by Panel B of Figure A2, among the second cohort of students, the baseline survey was administered in early October 2025. We organized a second “Organ Donation Awareness Week” from 27–31 October, following the same intervention protocol as in the spring.

2.3 Intervention

The treatment consists of an expert-led face-to-face intervention.⁹ CNPTO representatives visited the treatment classes during regular teaching activities in teams of two. The intervention started with a short video testimonial from a patient from rural north-western Tunisia, who had survived life-threatening illness thanks to a heart transplant performed by the CNPTO.¹⁰ We expected this component to reinforce the intervention: personal narratives have been found to be highly effective in changing attitudes (Oschatz and Marker; 2020). The video also illustrated that organ transplantation in Tunisia is managed by national institutions for the benefit of all citizens, helping to counter the widespread belief that organs can be bought if needed. Using a short powerpoint presentation, the experts then explained the medical procedure of organ transplantation. Moreover, they provided details on the legal framework around organ donation in Tunisia and explained to students how they could add the donor status to their ID. The presentation also included a few slides describing the Islamic perspective on organ donation. It concluded with a 10-minute Q&A session. Afterwards, the intervention team left the classroom and the students filled in the endline survey on their mobile device. Section E in the Online Appendix includes detailed in-

⁸The second round of the trial was not foreseen in the pre-analysis plan. We recruited 15 classes for the trial out of which 6 were randomly allocated to treatment and 9 to control, which increases the number of classes and students with respect to the pre-registered estimates.

⁹There is a growing literature on the effectiveness of expert- and layperson-led informational interventions with mixed results. Abu-Akel et al. (2021) show that the health messages pronounced by medical professionals during the COVID-19 pandemic are significantly more likely to be shared by the respondents than the messages pronounced by government officials or celebrities. Alsan and Eichmeyer (2024) provide evidence that expert-led health interventions are most effective among the *less* hesitant segments of the population. Evidence from a nationwide phone survey run by one of the authors (Hauser; 2025) confirms that the young and educated segments of the Tunisian population are relatively more willing to become organ donors. The choice of an expert-led intervention therefore appeared appropriate.

¹⁰This is the link to the video: <https://youtu.be/U6vYtpmkNeo>

structions for the intervention team. In the control group, the questionnaire was distributed by the lecturer on the same days as in the treated classes.

A short text at the end of the survey informed students that the CNPTO and the technical police would offer on-the-spot ID changes on campus on Thursday and Friday. This implies that students who considered adding their donor status to their ID had some time to think about their decision.¹¹ We followed the same protocol when we repeated the study in October 2025. All students — including those who had participated in the spring — were eligible to use the on-the-spot ID changes. The students who completed the survey received a lunch voucher for the university cafeteria, but there were no monetary or in-kind incentives for changing one’s ID. Appendix F includes some pictures from the intervention.

2.4 Theory of Change and Target Population

The intervention consists of two main components: (i) information about organ donation and (ii) real-life exposure to medical professionals. We are unable to disentangle the effects of these two components. We expected the treatment to increase students’ knowledge about organ donation as well as their trust in medical institutions. It may also alter their attitudes, social expectations and religious beliefs regarding organ donation. Better knowledge, higher trust, more progressive attitudes, social expectations and religious beliefs may eventually translate to policy-relevant behavioral change: more deceased organ donor registrations.

Our target population is interesting and relevant: from a psychological perspective, the opinions and attitudes of young people are generally more malleable than those of older individuals (Abrams; 2022). From a medical perspective, even though organs can be harvested from donors of all age groups, the transplants from young donors tend to be of better quality than those from older donors (Dayoub et al.; 2018). Importantly, even though overall mortality rates are lower, young adults represent a large — if not the largest — share of actual deceased organ donors.¹² Moreover, three out of the six Tunisian hospitals which are autho-

¹¹For ethical reasons, we had to give the students enough time to think about the decision to change their ID. However, we risked spillover effects from the treatment group to the control group as students from both treatment arms may interact. Within each round of data collection, we therefore left only a few days between the intervention and ID changes.

¹²According to Israni et al. (2024), the age group between 18 and 34 represented 27% of all deceased organ donors in the United States between 1988 and 2025, making it the most important age group in terms of absolute numbers. In the Eurotransplant area, 43% of all deceased organ donors were between the ages of 16 and 55 in 2024 (Eurotransplant International Foundation; 2025). Equivalent data is unfortunately not available for Tunisia.

rized to harvest and transplant organs are located in the capital (Zannad; 2024). Targeting a young urban population is therefore an efficient choice even from a welfare perspective.

2.5 Outcomes

Main Outcome: We treat ID changes as our primary outcome because it is a behavioral outcome and unlikely to be driven by experimenter demand effects.¹³ During the “Organ Donation Awareness Weeks”, students had the possibility to add their donor status to their ID on campus. In accordance with the Tunisian Law 99-27 of March 1, 1999, this required signing the official donor declaration form. The police agents then copied all relevant information for the new ID and took fingerprints. A photographer took biometric pictures on site. The whole procedure took less than 10 minutes. One week later, the CNPTO came back to the university to hand out the new IDs.¹⁴ In compliance with our Ethics protocol and the GDPR, we asked the students who had changed their ID if they were willing to share this decision with the research team, and they were given a printed privacy and consent form. All of the newly registered donors gave consent to share their data with us so this outcome is observed for all the students who decided to change their ID. We then matched this data with the administrative data on the students’ treatment status. The outcome is coded as 1 for students who changed their ID and 0 otherwise.

While we account for the on-the-spot ID changes in our intervention costs, they were available to all students and university staff members. The estimated treatment effect therefore describes the marginal effect of the expert-led informational sessions on top of removing logistical frictions related to ID changes. Due to the facilitation of ID changes, we expect the control group mean to be higher than the general population mean. This also implies that we are likely to underestimate the cost-effectiveness of the intervention.

Secondary outcomes: We measured six secondary outcomes with a Qualtrics survey

¹³The donor status on an ID represents a strong commitment to being a donor. It is possible to change one’s mind by going to a local police station and applying for a new ID. However, the process is costly in terms of time and effort.

¹⁴It is theoretically possible that students changed their ID in a local police station close to their home instead of making use of the ID changes we proposed on campus. However, the logistical costs associated with doing so are considerable, which is why we believe that this is unlikely to represent a threat to our measurement strategy.

tool: attitudes towards organ donation,¹⁵ social expectations,¹⁶ legal knowledge, medical knowledge, religious beliefs related to organ donation,¹⁷ and trust in medical institutions. The items are aggregated to standardized indices following [Anderson \(2008\)](#) for the main analysis, and we present disaggregated results in the Online Appendix. These outcomes, especially the attitudes, social expectations and trust measures, may be subject to experimenter demand effects. To test whether social desirability drives our findings, we collect data on students’ propensity to give socially desirable answers at baseline. Moreover, one of the social expectations questions is incentivized: students were asked to guess the number of peers who would register as organ donors during the awareness week. The student who guessed right was awarded a TND 50 lunch voucher.

In the six-month follow-up survey, we repeated most survey items to investigate whether treatment effects persist over time. Moreover, we asked students whether they had discussed organ donation with their family members and friends (Section [D.3](#)).

3 Empirical Analysis

We recruited a total of 1,277 students for the sample, 912 in the first round and 365 in the second round of the trial. Table [A1](#) shows the baseline balance of the full sample. Half of the students are female; a large share were born in Tunis. The only statistically significant imbalances are age (p-value 0.067), father’s education (p-value 0.079), and trust in the CNPTO, which is slightly lower in the treatment group (p-value 0.064). We therefore control for age, father’s education and baseline trust in the CNPTO in all main analyses.¹⁸ A randomization inference test of joint covariate orthogonality yields a p-value of 0.525, suggesting that randomization was successful. We present baseline balance for the first cohort in Table [A2](#) and the second cohort of students in Table [A3](#) respectively.

¹⁵Note that we dropped the question “I would like to be an organ donor myself” from the attitudes index due to its strong susceptibility to social desirability pressure after the pre-registration. We instead included the question “I worry that my family may disapprove of organ donation” in the index.

¹⁶Following [Bicchieri \(2006\)](#), we measure respondents’ (i) normative expectations about societal approval of organ donation, and (ii) their empirical expectations about peers’ organ-donor registration behavior.

¹⁷This outcome was added after the pre-registration.

¹⁸The share of parents with university education in our sample is very high and exceeds that of the general population by far. According to the [Barro and Lee \(2013\)](#) data, less than 9% of Tunisians aged 15-35 were enrolled in university in 2000. This likely reflects the fact that SMU is a private and highly selective university. The sampled students come from more educated backgrounds than the average Tunisian student.

3.1 Empirical Specification

To test the null hypothesis that the intervention had no impact on the outcome y_{ic} of student i in classroom c , we estimate a linear model using OLS:

$$y_{ic} = \alpha_0 + \alpha_1 T_c + X'_{ic} \gamma + \delta_s + \varepsilon_{ic} \quad (1)$$

where T_c is a binary variable which equals 1 if classroom c was allocated to treatment and 0 otherwise. X'_{ic} is a vector of control variables for student i in classroom c . We impute missing baseline covariates using the stratum-specific means or medians of the covariates. For network measures, we impute the stratum-and-treatment-group-specific mean or median. We control for the round of data collection, baseline trust in the organ donation center, student age, father’s education, the total number of social ties, and binary indicators for missing covariates in all our main regressions. We also include strata fixed effects δ_s , as the treatment was assigned stratifying by department and year of study.¹⁹ Standard errors are clustered at the classroom level. As a robustness check, we present all main results without covariate adjustment in Figure A4.

3.2 Spillover Analysis

The stable unit treatment value assumption (SUTVA) is unlikely to hold in our setting (Imbens and Rubin; 2015). As in similar informational interventions (i.e. Banerjee et al., 2024b), we expect spillovers to non-targeted individuals, as treated students may share the information from the intervention with friends in the control group. These spillover effects will bias $\hat{\alpha}_1$ towards 0. The baseline survey asked students to list the peers with whom they regularly discuss important issues, which allows us to compute the number of social ties to the treatment group for each student. We assume that, as social connectivity to the treatment group increases, a control group student is more likely to learn about the intervention from peers in the treatment group, thus being “contaminated” by spillovers.

We use the network data to estimate the following regression inspired by Miguel and Kremer (2004) and Baird et al. (2016), which assumes that spillovers increase linearly in the

¹⁹In our pre-analysis plan, we stated that we would use post-double selection (PDS) lasso to choose covariates. However, the lasso is unreliable in smaller sample sizes, which particularly affects the heterogeneity analysis and the follow-up survey analysis. We therefore opted for using a parsimonious set of baseline covariates throughout all analyses, which includes baseline variables that are slightly unbalanced or close to unbalanced, and highly predictive of the main outcome. Figure A3 shows the main results using PDS lasso.

number of treated social ties:

$$y_{ic} = \alpha_0 + \alpha_1 T_c + \alpha_2 N^T T_c + \alpha_3 N^T + \alpha_4 NT + \alpha_5 N + X'_{ic} \gamma + \delta_s + \varepsilon_{ic} \quad (2)$$

where N^T is the number of social ties in the treatment group, which, conditional on the total number of friends N , is exogenous. $\hat{\alpha}_3$ measures spillover effects — the impact of having more social ties to the treatment group. We thus explicitly allow for peer effects to differ by treatment arm while controlling for the total number of social ties. Table C1 reports social network measures from our baseline data collection, focusing on in-degree ties. The share of isolated students is balanced across treatment arms, as is the total number of social ties and the share of reciprocal ties. This suggests that the general structure of the social network is similar across treatment and control classrooms. As illustrated by Table C1 and Figure C1, the control group students have on average significantly more social connections among themselves than with the treatment group. Tables C2 and C3 show the same statistics by study round.

3.3 Follow-Up Analysis

The six-month follow-up survey was meant to provide additional evidence regarding the persistence of treatment effects on attitudes and spillover effects. Any treatment effects observed in the long run are unlikely to be driven by experimenter demand effects (Stantcheva; 2023). The response rate to the follow-up survey was very low, which is why we interpret these results with caution. Table A4 shows that 85.6% of invited students did not respond to the follow-up survey.²⁰ Attrition does not significantly differ between the two treatment arms (p-value 0.250). Yet, it is slightly lower in the treatment group (83.2% compared to 85.6% in the control group), which is why we present Lee (2009) bounds of the estimated treatment effects in Table A5. Table A4 shows that most observable baseline covariates, with the notable exception of gender in the whole sample, do not strongly predict attrition. Finally, Table A6 shows that the observable baseline characteristics of the students who answered the follow-up survey are fairly balanced across treatment arms. Keeping these caveats in

²⁰The low response rate is likely to be related to several factors: the baseline and the endline surveys were administered in the classroom, ensuring relatively high compliance. Six months later, the students were not necessarily taking courses with the same classmates and the follow-up survey was therefore sent to the students by the university’s research center. Some students had moved on to graduate studies, others had already graduated. Anecdotal evidence suggests that interest was low given that some students thought that they had already answered the same survey.

mind, we estimate inverse probability weights and add additional control variables, which predict attrition (self-reported baseline knowledge about organ donation, respondent gender, and an indicator for high religiosity), to all follow-up regressions.

4 Results

4.1 Treatment Effects on Donor Status

Table 1: ID Changes

	(1) Main	(2) Spillovers
Treatment	0.027** (0.010)	0.020** (0.010)
Treatment x N^T		-0.021 (0.013)
Treatment x N		0.006 (0.008)
N^T		0.020* (0.010)
N		-0.002 (0.003)
RI p-value	0.009	0.053
Control Mean	0.013	0.013
Observations	1277	1277

Notes: The table shows the estimated treatment effects on the main outcome, ID changes to add the donor status. All results are estimated using ordinary least squares (OLS) regressions. Column (1) shows the main specification as described in Equation (1). Column (2) shows results from the spillover effects regression using the number of treated students in a given student's social network (N^T) and its interaction with treatment as regressors as specified in Equation (2). All regressions control for student age, father's education, baseline trust in the organ donation center, the total number of social ties, the round of data collection, binary indicators for missing covariates, and strata fixed effects. Standard errors in parentheses are clustered at the class level. Randomization inference (RI) p-values (10,000 permutations) are reported in the last row. Inference accounts for clustering at the course level and stratification. Asterisks indicate significance at the *** 1%, ** 5% and *10% level.

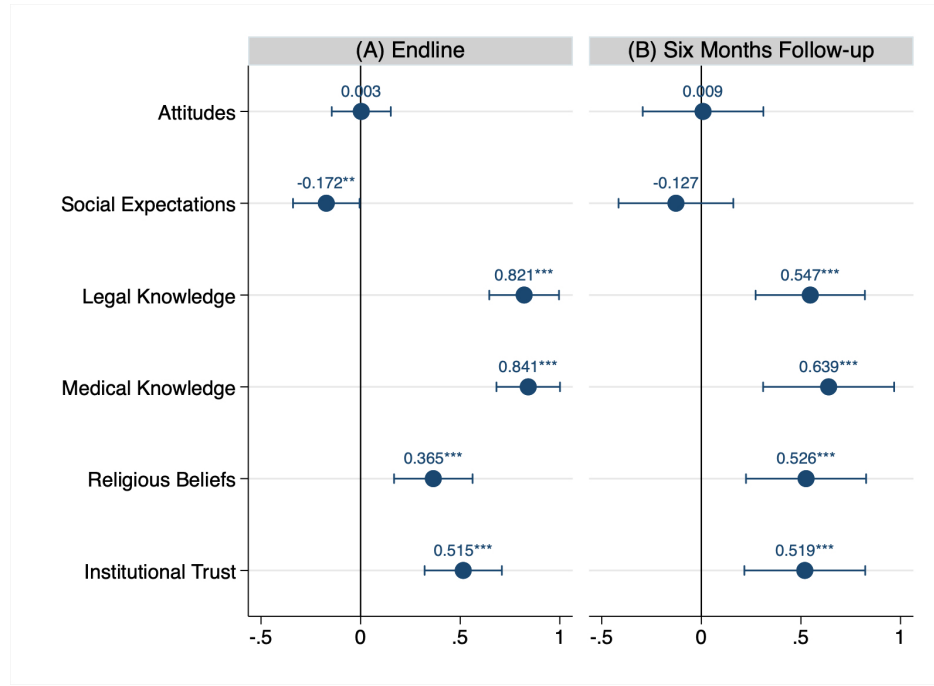
A total of 43 individuals decided to add the donor status to their ID during the two “Organ Donation Awareness Weeks” at SMU: 21 of them were students from the treatment group and 8 students from the control group. Moreover, 7 out-of-sample students and 7 staff members came to change their ID.²¹ Table 1 presents the estimated treatment effects on the students who were enrolled in the trial. Column (1) shows the result from estimating Equation (1): treated students are 2.7 percentage points more likely to change their ID. This corresponds to a 207% increase relative to the control group mean, as only 1.3% of control students changed their ID. Given that the control group mean is so low, the percent change should be interpreted with some caution. The estimated treatment effect decreases when the model is estimated without covariates in Figure A4, but stays statistically significant at the 5% level. Donor registration is a rare outcome and the treatment effect estimate may be affected by a small number of control students who decide to register. Table A7 shows that ID changes were spread across classes and occurred in 13 treatment classrooms compared to only 7 control classrooms. We also report Fisher exact p-values using the clustered treatment assignment in Table 1. Randomization inference yields a p-value of 0.009, meaning that out of 10,000 permutations, only 90 produced an effect size greater than or equal to the one observed.

Column (2) of Table 1 provides some evidence in favor of our initial hypothesis of positive spillover effects to the control group. When estimating Equation (2), we find that $\hat{\alpha}_3$ is positive and marginally statistically significant at the 10% level: if we were to increase the number of treated students among a control student’s social network from 0 to 1, all else equal, we would expect them to become 2 percentage points more likely to register as a deceased organ donor. Table C4 shows the same estimates by study round. The follow-up survey asked students whether they had discussed organ donation with their friends. Table A8 shows that treated students are marginally more likely to report having discussed organ donation with their friends, corroborating this result.

As a robustness check, we estimate the same equation restricting to social ties which are reciprocal. Reciprocal ties imply that two students nominated each other, which is arguably a stronger measure of connection than a mono-directional tie. Yet, by restricting to reciprocal ties, we effectively reduce the number of observations across which we estimate spillover effects, as only a fraction of all social ties are reciprocal. The estimated spillover effects lose

²¹The 7 staff members who changed their ID include 2 professors who taught the treatment classrooms, 2 staff members who were involved with the Organ Donation Awareness Week as they worked in communications, and 3 not directly related staff members.

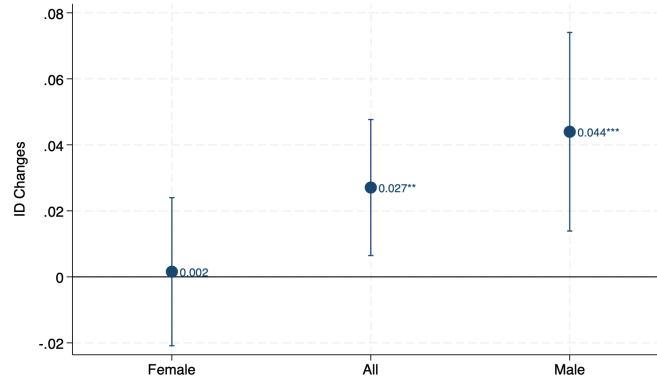
Figure 1: Mechanisms



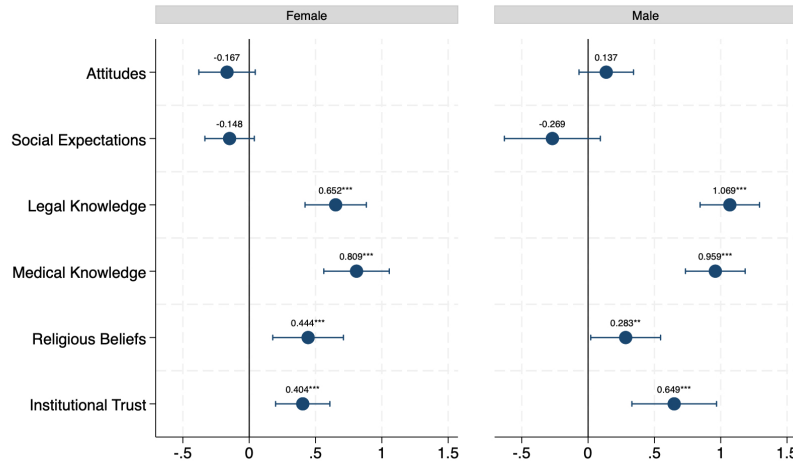
Notes: The figure plots the estimated treatment effects on the secondary outcomes, aggregated to indices. The indices are standardized to have mean 0 and standard deviation 1 in the control group. All results are estimated using ordinary least squares (OLS) regressions. Positive values indicate more progressive attitudes, norms and religious beliefs, increases in knowledge and institutional trust. See Section D.2 for a full list of items by index. All regressions in Panel (A) control for the baseline outcome if available, baseline trust in the organ donation center, student age, father's education, the total number of social ties, the round of data collection, binary indicators for missing covariates, and strata fixed effects. The set of controls for the regressions in Panel (B) includes the baseline outcome, if available, baseline trust in the organ donation center, student age, father's education, having no social ties, self-reported baseline knowledge about organ donation, respondent gender, an indicator for high religiosity, the round of data collection, binary indicators for missing covariates, as well as strata fixed effects. The regressions in Panel (B) are weighted using inverse probability weights. Standard errors are clustered at the class level. Asterisks indicate significance at the *** 1%, ** 5% and *10% level.

Figure 2: Heterogeneity by Gender

(a) ID Changes



(b) Mechanisms



Notes: The figure plots the estimated treatment effects on ID changes (Panel A) and the secondary outcomes aggregated to indices (Panel B) by gender. All results are estimated using ordinary least squares (OLS) regressions, we estimate the model separately for female and male respondents. The indices are standardized to have mean 0 and standard deviation 1 in the control group. Positive values indicate more progressive attitudes, norms and religious beliefs, increases in knowledge and institutional trust. See Section D.2 for a full list of items by index. All regressions control for the baseline outcome if available, baseline trust in the organ donation center, student age, father's education, the total number of social ties, the round of data collection, binary indicators for missing covariates, and strata fixed effects. Standard errors are clustered at the class level. Asterisks indicate significance at the *** 1%, ** 5% and *10% level.

was entirely female — a role model effect should have favored registration among female students (Beaman et al.; 2012; Porter and Serra; 2020). A priori, we therefore expected larger treatment effects on donor registration among women. Instead, we find that 22 out of the 29 students who registered as organ donors (76%) were male. Figure 2 reveals that the estimated treatment effects on donor registration are entirely driven by young men. Male

the possibility of peer-to-peer learning, but they are imprecisely estimated. Even though women exhibit comparable gains in knowledge and institutional trust, treatment effects are driven by young men, which is at odds with women’s baseline intentions. This discrepancy and gender-specific treatment effects on attitudes are consistent with women having less agency to act on their preferences than men. Donor registration is higher among students from supportive families, adding evidence to the hypothesis that family dynamics play an important role. Given that organ donation seems to be viewed as a collective decision in this context, the intervention’s impact might be increased by involving both peer networks and older family members.

While Tunisia’s deceased organ donor rates are similar to those of many LMICs in the MENA region and beyond, the study was conducted in the setting of a private university, among students who are not representative of the general population. Yet, this group remains relevant and interesting, as young adults are more sensitive to new information and may spread it among their social network. Targeting this age group is efficient from a medical perspective since the organs donated by young donors are, on average, of better quality than those of older donors. Most importantly, despite overall lower mortality rates, young adults tend to be overrepresented among actual deceased organ donors. We therefore believe that the findings of our study can inform health policy in Tunisia and in similar contexts. Future research should expand the target group and explore intra-family decision-making dynamics in greater depth.

Acknowledgements

This research was only possible thanks to the intellectual and practical support of the Tunisian Center for the Promotion of Organ Transplantation (CNPTO). We would like to express our gratitude to the CNPTO and especially to Dr. Boutheina Zannad, Dr. Inès Jaafar, Khadouja Ben Nakissa and Mejda Chaouachi. We also thank Leila Triki, Sana Mami, Imen Bouhestine, Lamia Ben Fdhila, Emnaa Chahed, and the faculty of the South Mediterranean University. Finally, we thank Sule Alan, Pietro Biroli, Juan José Dolado, Guido Friebel, Nicola Lacetera, Isabel Rodríguez, Jaromir Sant and Alessandro Tarozi for helpful comments.

Skill certifications and Migrant Labor Market Integration: Experimental evidence from Colombia

September 18, 2026

Abstract

This paper studies whether government-issued skill certification improves labor market integration when workers' skills are hard to verify. We evaluate a randomized offer of access to an occupation-specific certification program in Colombia among locals and migrants, two groups with different ex ante signaling capacity. The program certifies existing skills rather than providing training. Among locals, the offer increases employment by reducing inactivity and raises hours worked. The effects persist unevenly over time and are robust to attrition bounds and alternative outcome measures. Among migrants, estimates are less precise and provide no evidence of effects in employment, hours, or wages. The results show that a common skill certification generates different patterns of labor-market adjustment among locals and migrants.

Keywords: Signaling, Certifications, Skills, Migration, Colombia

JEL: J24, J61, J64, I26, O15, D82.

1 Introduction

Migrants often enter destination labor markets with education and work experience that are less familiar to local employers. This can make their skills harder for employers to assess and can limit the value of the signals they bring from their country of origin. As a result, migrants may have difficulty finding stable employment, work in jobs that do not match their qualifications ([Dustmann et al., 2012](#); [Lebow, 2023](#)), or remain concentrated in informal employment ([Delgado-Prieto, 2025](#)). These problems can persist even after legal barriers to employment are removed ([Bahar et al., 2021](#); [Ibáñez et al., 2024](#)). More generally, when employers have limited information about workers' skills, job seekers rely on signals such as degrees, completed courses, grades, and institutional reputation ([Spence, 1973](#)).

This paper studies whether a credible skill signal improves labor market integration. We focus on skill certification, a policy that verifies existing competencies rather than providing training. We estimate the effects of an offer of access to an occupation-specific certification program for both locals and migrants. This design allows us to test whether the same signal affects workers with different ex ante signaling capacity along different labor market margins.¹

We study a free certification program implemented by the Colombian Ministry of Labor. The program certifies occupation-specific skills acquired through work experience or on-the-job training rather than formal education. Eligibility requires applicants to document prior work experience in the occupation linked to the certification. Among 2,160 eligible applicants, 855 randomly received an offer of access to the program, including 250 migrants. Treatment-assigned applicants were offered competency assessments and, if successful, certification. Control-group applicants were not assessed and did not receive certification during the evaluation period. Treatment and control applicants were surveyed at three, six, and twelve months after the intervention.

We estimate intention-to-treat effects separately for locals and migrants. Random assignment produced balanced treatment and control groups within both samples. Survey response rates are high, but treatment assignment increased follow-up response in some waves, especially among migrants. We report Lee bounds ([Lee, 2009](#)) to assess the sensitivity of the main estimates to differential attrition.

The effects differ across groups. Among locals, the offer of access to the certification program increases employment by 5.7 percentage points and reduces inactivity by 3.5 percentage points. It also raises weekly hours worked by 2.0 hours and full-time work by 3.7 percentage points, but does not increase formal employment or wages among

¹The study was conducted under a pre-analysis plan accepted as part of the registered-report process, which specified the randomized offer of access to the certification program, separate analyses for locals and migrants, primary labor-market outcomes, and analyses of channels.

those already employed. Among migrants, the estimates are less precise, and we do not detect improvements in employment, hours worked, or wages. The migrant point estimates are smaller in magnitude than the minimum detectable effects implied by the design, indicating limited statistical power to detect effects of this size—and cautioning against interpreting the null results as evidence of zero effects.

Dynamic estimates show that the local employment effect emerges quickly and remains positive after one year, while the migrant estimates remain imprecise. The difference in employment effects between locals and migrants is statistically significant. Although the two groups differ along baseline characteristics, reweighting the migrant sample to match locals on observables leaves the heterogeneous pattern unchanged. These results suggest that observable differences in sample composition do not fully account for the contrasting estimates, but the experiment does not identify why the effects differ across groups.

The evidence on additional outcomes provides some guidance on potential channels but does not identify them directly. For locals, the program reduces unemployment duration by 0.23 months, without detectable changes in reservation wages or search intensity. This pattern is consistent with certification reducing screening frictions and facilitating labor market entry. For migrants, we find no clear pattern in job-search outcomes that explains the main estimates. We also find little evidence that changes in education or training enrollment explain the effects for either group over the period we study.

Compliance was high but not perfect. A small share of participants randomized into treatment did not receive a certificate because they did not successfully complete the evaluation process or because they withdrew. We therefore examine whether the intention-to-treat estimates are sensitive to incomplete compliance by instrumenting receipt of certification with random assignment to treatment. The instrumental variable estimates closely mirror the intention-to-treat estimates, as expected given the high compliance rate. Stated willingness to participate in the program without subsidies is high in both treatment and control groups, suggesting that participants valued the program regardless of assignment.

This paper contributes to the literature on labor market signaling. This literature shows that credible information about workers' skills can affect hiring and labor market outcomes. Experimental studies find positive effects from certification or information disclosure among young job seekers in Ethiopia ([Abebe et al., 2021](#)), during job interviews in Uganda ([Bassi and Nansamba, 2022](#)), among young job seekers in South Africa ([Carranza et al., 2022](#)), among young workers in New York ([Heller and Kessler, 2021](#)), and in online labor markets ([Pallais, 2014](#)). Related evidence shows that labor market signals such as grade point averages, college reputation, institutional quality, and

publicly available skills information can generate substantial returns (Landaud et al., 2024; MacLeod et al., 2017; Eble and Hu, 2022; Busso et al., 2025b). We contribute by testing whether a common, occupation-specific skill certificate generates heterogeneous returns for two groups with different ex ante signaling capacity, locals and migrants. We show that this certification produces different patterns of labor-market adjustment across the two groups.

The study also contributes to the literature on migrant labor market integration. This literature analyzes the effects of policies that reduce barriers to employment in destination labor markets, including job search assistance (Battisti et al., 2019), intensive coaching (Joona and Nekby, 2012), wage subsidies, training, and placement programs (Dahlberg et al., 2024; Foged and Van der Werf, 2023; Foged et al., 2024), citizenship application subsidies (Hainmueller et al., 2018), mentorship programs (Jaschke et al., 2022), refugee-firm matching (Loiacono and Silva-Vargas, 2023), and regularization policies (Bahar et al., 2021; Ibáñez et al., 2024). These interventions can improve employment, earnings, legal status, or social integration, but evidence from Colombia suggests that regularization alone had limited effects on the labor market integration of Venezuelan migrants, especially among lower-skilled workers (Bahar et al., 2021; Ibáñez et al., 2024).

We contribute by studying a policy that targets a different barrier to migrant integration: the difficulty employers may face in assessing skills acquired abroad. Unlike training or intensive job-search programs, skill certification does not aim to build new skills. It verifies existing competencies and provides a domestic credential. Other forms of recognition, such as the validation of foreign degrees or professional credentials, can be costly and time-consuming. In their absence, migrants may retrain unnecessarily, generating productivity losses (Carranza and McKenzie, 2024a). Our results show that the effects of this type of intervention need not be the same for migrants and locals. While certification increases employment among locals, the estimates for migrants are less precise and do not show positive effects on labor market outcomes.

The rest of the paper is organized as follows. Section 2 describes the program setting and intervention. Section 3 presents the research design, data, outcomes, and empirical strategy. Section 4 assesses internal validity, including balance and attrition. Section 5 reports the main results and their dynamics. Section 6 presents robustness checks. Section 7 discusses certification receipt. Section 8 examines channels and additional outcomes. Section 9 presents a policy discussion, and Section 10 concludes.

2 Program Setting and Intervention

Colombia provides a useful setting to study whether certification can make existing skills more legible to employers. The country received a large inflow of Venezuelan migrants over a short period, many of whom obtained legal status and could participate in the labor market.² Despite their higher labor force participation, Venezuelan migrants face substantial occupational mismatch (Kaplan et al., 2023). Credential non-recognition and employer uncertainty about migrant qualifications push many into informal or low-skilled jobs below their competency level (Bahar et al., 2021; Lebow, 2023). Colombia also has a long-standing system of skills certification, allowing us to study whether an occupation-specific domestic signal affects labor market outcomes.

2.1 Skill Certification Program

Since 2003, the Colombian government has operated a free skill certification program through the Colombian Vocational Training Agency (SENA).³ The program allows adults with prior work experience to validate existing occupational skills through publicly recognized certifications. It does not provide training. Instead, the certifying agency assesses whether applicants have already acquired specific competencies through work experience.⁴

The certifications can credibly signal workers' skills to Colombian employers for three reasons. First, the program certifies more than 200,000 workers each year throughout the country, making the credential widely used. Second, SENA is a recognized national institution that provides both certification and technical training, although these services are administered separately. Third, certification assessments are based on occupational standards developed with input from industry associations, giving employers a role in defining the competencies being certified. These features make the certificate a potentially informative signal of occupation-specific skills.

Eligibility requires a valid identification document, registration through the program website, acceptance of the legal terms, and evidence of at least six months of prior work experience in the occupation linked to the certification. The program offers assessments in 2,405 competencies across manufacturing, agriculture, retail, and service occupations. Applicants may be employed or unemployed, but they must select a specific competency for certification and document prior practical experience in the corresponding occupation.

²See Appendix A for details on the Venezuelan migration crisis, the regularization process, and migrants' labor market integration in Colombia and across the region

³The program is named, in Spanish, "Programa de Evaluación y Certificación de Competencias Laborales", which was regulated by Decree 933 of 2003.

⁴Appendix B provides additional details about the skills certification program.

Candidates are evaluated through a practical assessment and a knowledge test. In the practical assessment, a qualified assessor from the certifying agency evaluates whether the applicant can perform occupation-specific tasks and whether they meet the competency standards, assigning a pass/fail grade. Applicants who pass this assessment then complete a multiple-choice knowledge test administered virtually, with scores ranging from 0 to 100. Successful candidates receive a certificate that reports their assessed level: basic, intermediate, or advanced.⁵ Candidates who do not pass are eligible for a six-hour remedial session and may subsequently retake the assessment. Historically, fewer than 10 percent of participants fail, consistent with the requirement that applicants document prior practical experience in the occupation before certification.

2.2 “Skills Pay Off” Program

Since 2021, *Saber Hacer Vale* (“Skills Pay Off”) has extended access to the certification program to vulnerable populations. Implementation begins with the selection of a geographic area and the competencies to be certified. The agency then determines the number of available slots, opens the call for applications, and verifies applicants’ documentation to determine eligibility.⁶

Certification is the central component of the intervention, but the treatment also includes two additional components: information on publicly available job-search resources and financial assistance during participation.⁷ The information resources are unlikely to drive any treatment effects, as they are publicly available to both the treatment and control groups, participation is voluntary, and they constitute light-touch interventions (Carranza et al., 2022; McKenzie, 2017; Kelley et al., 2024).

The financial assistance includes food and connectivity subsidies of approximately \$82 USD for all participants, with additional transportation, dependent-care, or remedial-session support for those who qualify. Total support can reach \$217 USD, equivalent to roughly 70 percent of the monthly minimum wage. Because benefits are paid upon completion, the payment structure creates incentives to start and complete the process. The transfers were intended both to compensate participants for the opportunity cost of participation and to relax access constraints, including virtual participation and childcare costs, which program officials identified as particularly salient for certain

⁵The type of certificate awarded depends on the grade obtained in the evaluation process: those with scores below 30 fail the exam and do not receive the certificate; those with scores between 30 and 60 receive a basic certificate; those between 60 and 90 receive an intermediate certificate; and those above 90 receive an advanced certificate. The issued certificate explicitly states the level at which the individual was classified.

⁶Appendix B provides additional details about “Skills Pay Off”.

⁷Appendix B provides a detailed description of both additional components of the program.

demographic groups, such as migrant women. However, they could also relax liquidity constraints after certification. In this sense, the program is best understood as a certification program that reduces the cost of participation for low-income individuals.

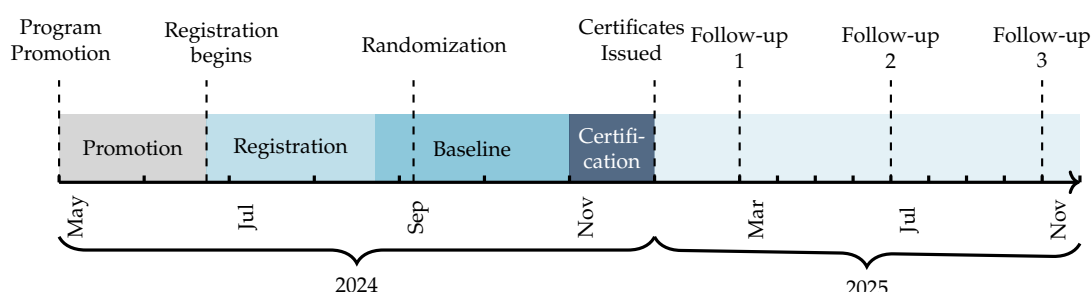
Eligibility for additional support requires meeting at least one vulnerability criterion, such as low income, caregiving responsibilities, age, ethnic-minority status, or being a migrant. At least 30 percent of available slots are reserved for migrants with regular status.

2.3 Intervention

The intervention evaluated in this paper is the 2024 cohort of “Skills Pay Off,” implemented in Colombia’s South-western region.⁸ The cohort had 855 available slots across five competencies: health services, food preparation, food handling, design and tailoring, and recycling operations.

Figure 1 summarizes the implementation and evaluation timeline. The program was advertised in May and June 2024, and applications were open between June and August 2024. A total of 2,736 individuals registered through the online application form. Of these, 2,160 met the eligibility criteria, which primarily verified prior work experience in the occupation linked to the requested certification.⁹

Figure 1: Timeline of the Intervention



Applicants offered access to the program completed the standard certification process, which included a practical assessment administered by a qualified evaluator and a multiple-choice knowledge test from the certifying agency. Applicants who met the competency standards received an official certificate indicating their assessed skill level. Applicants who did not initially meet the competency standards were offered a remedial session and could retake the assessment.¹⁰ Certificates were issued between November and December 2024.

⁸Specifically, in the departments of Valle del Cauca, Cauca, and Nariño.

⁹Submitted documents were reviewed between August 1st and September 3rd to determine eligibility for the program.

¹⁰Participants who completed the remedial session and subsequently passed the assessment received the certification and the corresponding financial incentives.

Table 1 reports certification outcomes among applicants assigned to treatment. Compliance was high but incomplete: 89 percent of treatment-assigned applicants received a certificate. Certification levels were 15 percent basic, 23 percent intermediate, and 51 percent advanced. The remaining 11 percent did not receive a certificate: four applicants were assessed but did not meet the certification criteria, while 89 withdrew before completing the assessment process.

Table 1: Certification Assessment Outcomes and Treatment Assignment by Migration Status

Treatment Status	Local Sample		Migrant Sample	
	(n)	(%)	(n)	(%)
Advanced	333	19.9	104	21.5
Intermediate	114	6.8	84	17.4
Basic	87	5.2	40	8.3
Not competent	3	0.2	1	0.2
Withdrawn	68	4.1	21	4.3
Control	1,071	63.9	234	48.3

Notes: The table reports the distribution of participants by migration status. Advanced, Intermediate, Basic, and Not Competent refer to the proficiency level at which each treated participant was classified after completing the evaluation and certification process. Withdrawn denotes individuals assigned to treatment who did not complete the process. Control denotes individuals randomly assigned to the control group.

3 Research design

3.1 Randomization

Before 2024, access to the program was allocated on a first-come, first-served basis. In the 2024 cohort studied in this paper, excess demand led the Ministry of Labor to allocate access by lottery. On September 6th, 2024, the 855 available slots were randomly allocated among the 2,160 eligible applicants.

Randomization was conducted at the individual level, stratified by type of certification (i.e., competency) and whether applicants were migrants, with 30 percent of slots reserved for migrants. Individuals assigned to treatment were offered access to the certification program as implemented.

Individuals assigned to control were informed that they would be offered access to the certification program in a later round, approximately 12 months after the intervention began.¹¹ This could have discouraged them from pursuing alternative certifications during the evaluation period even when, absent that promise, they would have done so. Such anticipation effects could mechanically widen treatment-control differences

¹¹The randomized allocation had to satisfy the Ministry of Labor's budget constraint.

and bias our estimates upward. To assess this concern, we use administrative records from earlier *Skills Pay Off* calls, in which some eligible applicants were not certified because available slots were limited. About 4.3 percent of these eligible non-participants obtained a certification within one calendar year of being rejected, compared with 5.2 percent of individuals in the randomized control group over the corresponding period. We cannot reject equality between the two rates. These results provide no evidence that anticipated future access materially changed the behavior of the control group.¹²

3.2 Data Collection

Participants assigned to treatment and control were surveyed before and after the intervention. Baseline data come from two sources. The registration form collected demographic, socioeconomic, educational, and employment characteristics at the time of application. Before the intervention, participants were also contacted by phone in September 2024 for a complementary baseline survey, which collected additional information on labor market outcomes, mental health, and poverty.

We then conducted three follow-up surveys, at three, six, and twelve months after the program ended. In each round, participants were recontacted by phone and invited to complete a survey comparable in scope to the baseline. The first follow-up took place between February and March 2025; the timing of all survey rounds is shown in Figure 1.¹³

3.3 Pre-Analysis Plan

Our analysis follows a pre-analysis plan that specifies randomized access to the certification program as the treatment and stipulates separate intention-to-treat analyses for locals and migrants (Busso et al., 2025a). It considers effects on employment and wages and allows these effects to differ between locals and migrants.

The pre-analysis plan also lays out several potential channels through which certification could affect labor-market outcomes. On the employer side, certification may improve screening by making workers' skills easier to assess. On the worker side, it may change job-search behavior and the types of offers workers are willing to accept. The plan also considers subsequent education and training responses. These outcomes do not provide direct tests of any one mechanism, since the experiment

¹²See Appendix B.5 for additional details. Self-reported certification in the follow-up surveys also closely matches the administrative records for the randomized control group.

¹³The pre-analysis plan anticipated a nine-month follow-up survey. This round was not collected because of budgetary and implementation-timing constraints. Although the first follow-up survey was fielded before the pre-analysis plan was accepted, the research team did not receive any follow-up data until after acceptance.

neither manipulates nor directly observes employer beliefs or workers' underlying constraints. We therefore use them to assess whether the pattern of results is consistent with these different channels. The plan also identifies skill downgrading as an outcome of particular interest for migrants, reflecting the possibility that certification improves occupational allocation.

3.4 Outcome Measures

Following the pre-analysis plan, we examine labor-market outcomes together with reservation wages, unemployment duration, search intensity, and subsequent enrollment in education, certification, or training. The latter outcomes help interpret the main labor-market effects but do not identify specific mechanisms. In particular, they allow us to assess whether the pattern of results is consistent with improved employer screening, changes in workers' job-search behavior, or subsequent education and training responses.

The labor-market outcomes capture employment status and job quality. We measure whether respondents are employed, unemployed, or economically inactive, where inactivity is defined as not working and not actively seeking work. We also measure monthly wages, weekly hours worked, full-time work, and formal employment, defined as employment in which the worker contributes to the pension system. All these outcomes are measured directly in the surveys.

We measure *skill downgrading*, as an indicator equals to one for individuals with post-secondary education who are employed in occupations requiring fewer qualifications than their educational level, and zero otherwise. To construct the measure, we use the Colombian Classification of Occupations (CUOC),¹⁴ which maps each occupation to a skill level based on the complexity and qualifications required to perform its tasks.

To study potential mechanisms, we collected two sets of additional outcomes. The first captures job-search behavior: whether the respondent received a job offer, the number of offers received, weekly hours devoted to job search, unemployment duration, and reservation wages. Reservation wages are elicited separately for employed and unemployed respondents as the minimum salary at which they would switch jobs or accept a job offer, respectively. The second set captures post-program education and training.

We also collected secondary outcomes on job satisfaction, well-being, and program valuation. Job satisfaction is measured by an indicator for reporting being satisfied or very satisfied with the current job. Well-being is measured by anxiety, using the GAD-7

¹⁴The CUOC adapts the International Standard Classification of Occupations (ISCO-08) to the Colombian context.

scale, and food insecurity, using an indicator for having skipped a meal because of lack of economic resources. Program valuation is measured by respondents' willingness to pay for a similar skill certificate and willingness to participate in the program without subsidies.¹⁵

3.5 Empirical strategy

The primary estimand is the intention-to-treat effect (ITT) of being offered access to the certification program as implemented. The main specification pools the three post-treatment survey rounds:

$$y_{ijt} = \alpha_0 + \alpha_1 D_i + \mu_j + \mu_t + \varepsilon_{ijt}, \quad \text{for } M_i \in \{0, 1\}, \quad (1)$$

where y_{ijt} represents an outcome for individual i , enrolled in certification area j , in survey wave t ; D_i is an indicator equal to one if the individual was randomly assigned to receive an offer of access to the certification program; M_i is an indicator equal to one for migrants; and μ_j and μ_t denote competency and survey-wave fixed effects, respectively. The coefficient α_1 identifies the average intention-to-treat effect for the corresponding subgroup.

Because not all treatment-assigned applicants received a certificate, we also estimate IV effects of certification receipt. We instrument receipt of certification with random assignment to treatment. The resulting 2SLS estimand is a local average treatment effect (LATE) for compliers.

We also estimate time-specific effects to examine whether the impact of the offer of access to the certification program varies over time. This allows us to assess timing and persistence. The dynamic specification takes the form:

$$y_{ijt} = \beta_0 + \beta_1 D_i + \sum_{t>T_0}^T \beta_{2t} (D_i \times T_t) + \mu_j + \mu_t + \varepsilon_{ijt}, \quad \text{for } M_i \in \{0, 1\}, \quad (2)$$

where all terms are defined as in Equation (1). The indicators T_t denote follow-up survey waves. The coefficients β_{2t} capture how the treatment effect differs across follow-up waves relative to the omitted survey wave. In this specification, each outcome is measured contemporaneously at each follow-up wave and reflects the respondent's status at that point in time (i.e., outcomes are not carried forward across waves). A respondent's value at a later follow-up is determined by their situation at that survey, regardless of their status in earlier waves. Respondents who do not complete a given

¹⁵Appendix C defines the outcome measures used in the paper, maps them to the pre-analysis plan, documents planned outcomes not reported in the final analysis, and describes the construction of derived outcomes, including skill downgrading.

follow-up are excluded from the estimate for that wave.

Equations (1) and (2) are estimated separately for locals and migrants. These subgroup estimates are internally valid because randomization was stratified by whether applicants were migrants. For completeness, we also report pooled interacted specifications and tests of equality between local and migrant effects. However, differences in estimated effects across groups should be interpreted with caution. Locals and migrants differ in pre-treatment characteristics, and these characteristics may moderate the returns to certification. As a result, cross-group differences in treatment effects may reflect migration background, observable or unobservable moderators, or some combination of these factors. The design does not separately identify these channels.

In all cases, standard errors are clustered at the individual level. Because we estimate effects on multiple outcomes, we adjust inference using sharpened q-values following [Anderson \(2008\)](#). We compute q-values within outcome families in each table and separately by estimation sample. Standard errors are reported in parentheses, while q-values provide the multiple-testing adjustment used to interpret significance across related outcomes.

4 Internal Validity

4.1 Balance

We assess randomization balance by comparing baseline characteristics between treatment and control applicants separately for locals and migrants. The tests condition on competency fixed effects, reflecting the stratified randomization design. For descriptive purposes, we also compare locals and migrants.¹⁶

Treatment and control applicants are balanced across observable baseline characteristics within both samples. Only one test rejects equality at the five-percent level, a pattern consistent with chance given the number of characteristics tested. This supports treatment-control comparisons within the local and migrant samples.

Locals and migrants differ along several baseline dimensions. Migrants are younger, more educated, more likely to be employed, and more likely to experience skill downgrading at baseline. These differences do not threaten identification within each subgroup, but they imply that cross-group differences in treatment effects cannot be attributed mechanically to migrant background alone. They also motivate the paper's focus on skill downgrading as a central margin of migrant labor market integration. Migrants are more educated—18 percent hold university or postgraduate degrees

¹⁶Baseline balance tests are reported in Appendix [D](#).

compared to 8 percent of locals—yet report similar wages. Among migrants with professional education, 12 percent work in occupations requiring fewer skills than their qualifications, compared to 4 percent of locals.¹⁷

4.2 Survey Attrition

Attrition can be an important threat to internal validity after random assignment. Recontacting applicants after registration is difficult, and treatment assignment may affect survey response because treated applicants remain connected to the program while control applicants do not. To reduce non-response, all participants were offered monetary incentives for completing follow-up surveys, independent of treatment status.¹⁸ These incentives were designed to preserve engagement in both the treatment and control groups.

Overall response rates are high. About 90 percent of participants completed the complementary baseline, and more than 82 percent completed the twelve-month follow-up.¹⁹ However, response rates differ by treatment status in some waves. Differential attrition is moderate and concentrated among migrants: treatment-assigned migrants are more likely to respond at the three- and six-month follow-ups and, more weakly, at twelve months.

To assess the sensitivity of the estimates to differential attrition, we report bounds following [Lee \(2009\)](#). The approach trims the outcome distribution of the group with the higher response rate in proportion to the response-rate gap between treatment and control. It relies on a monotonicity assumption: treatment assignment weakly increases the probability of survey response. This assumption is plausible in our setting because treatment-assigned applicants remained connected to the program, while control applicants had no comparable program-related reason to respond. As a robustness check, we report the resulting bounds and discuss their implications for the main estimates.

¹⁷This pattern is less pronounced for the broader skill-downgrading measure that includes individuals with technical education, which we attribute to limited occupational detail at baseline.

¹⁸Participants received \$7 USD for each completed survey and could receive up to \$30 USD for completing all follow-up rounds. Participants who missed a round could still participate in later rounds and remain eligible for the full incentive if they completed the required survey modules.

¹⁹Response rates by treatment status and nationality, along with tests of differential attrition by migrant background and wave, are reported in Appendix E. Among migrants, differences are statistically significant at the 1 percent level in the three-month follow-up wave, at the 5 percent level in the six-month wave, and at the 10 percent level in the twelve-month wave. Among locals, only the three-month difference is significant, at the 10 percent level.

5 Results

5.1 Main Results

Table 2 reports intention-to-treat estimates from Equation (1). The estimates pool observations across the three follow-up rounds and are reported for the full sample (Panel A), locals (Panel B), and migrants (Panel C). All specifications include competency and survey-wave fixed effects.

In the full sample, access to the certification program increases employment by 3.6 percentage points, a 7 percent increase relative to the control mean. The effect on the zero-imputed log wage is positive, consistent with the employment response. Conditional estimates restricted to employed workers show no corresponding wage gain, indicating that the earnings effect is driven by the extensive margin rather than by higher wages among those employed (-0.020 , $p\text{-value}=0.650$). We find no statistically significant effects on formal employment, weekly hours worked, full-time work, or skill downgrading in the full sample.

Among locals, access to the certification program increases employment by 5.7 percentage points and reduces inactivity by 3.5 percentage points, suggesting that part of the employment gain reflects transitions out of inactivity. Certifications also increase the zero-imputed log wage, weekly hours worked by 2.0 hours, and the probability of full-time work by 3.7 percentage points. The hours effect represents a 15.7 percent increase relative to the control mean reported in Table 2.²⁰

The positive effect on the zero-imputed wage is mechanical, reflecting more locals earning a wage. Among those employed at follow-up, the effect on the log wage is small and not statistically significant (0.037 , $p\text{-value}=0.810$). This pattern runs against our original hypothesis that informative signals would generate wage returns alongside employment effects. This latter effect should be read with caution, however, because it conditions on employment at follow-up (itself an outcome affected by treatment). Workers who enter employment because of the offer are disproportionately drawn from those unemployed or inactive at baseline. If these marginal entrants earn less than workers who would have been employed regardless of treatment, their entry would lower average wages among treated workers. This selection could push the conditional wage estimate toward zero even if certification increases wages for a given match. The null conditional wage estimate should therefore not be interpreted as evidence of no wage returns to certification.²¹

²⁰For locals, the employment, inactivity, log-wage, weekly-hours, and full-time effects retain significance at conventional levels after the multiple-testing adjustment (q-values below 0.05). The full-sample employment and log-wage effects do not survive the adjustment (q-values of 0.192).

²¹Beyond this selection concern, two features of this labor market may also limit wage responses,

Table 2: Effects of Access to Skill Certification on Labor Market Outcomes

Intention To Treat estimated using Equation (1)

			Outcomes Replaced by Zero if Non-Employed				
	Employed (1)	Inactive (2)	Log Wage (3)	Formal emp. (4)	Weekly working hours (5)	Full time (7+ hrs) (6)	Skill Downgrade (7)
<i>Panel A: Full Sample</i>							
Treatment	0.036** (0.018)	-0.024 (0.015)	0.490** (0.245)	-0.001 (0.013)	1.223 (0.784)	0.024 (0.016)	-0.016 (0.015)
Q-value	0.192	0.192	0.192	0.378	0.192	0.192	0.227
Mean Control	0.530	0.240	6.734	0.128	14.75	0.256	0.156
MDE	±0.050	±0.042	±0.686	±0.037	±2.197	±0.045	±0.042
Observations	5647	5647	5634	5647	5647	5647	5647
<i>Panel B: Locals</i>							
Treatment	0.057*** (0.021)	-0.035* (0.018)	0.741*** (0.283)	-0.008 (0.015)	2.016** (0.857)	0.037** (0.017)	0.004 (0.016)
Q-value	0.032	0.046	0.032	0.258	0.033	0.042	0.293
Mean Control	0.494	0.264	6.199	0.136	12.88	0.221	0.136
MDE	±0.059	±0.050	±0.792	±0.043	±2.401	±0.049	±0.045
Observations	4404	4404	4392	4404	4404	4404	4404
<i>Panel C: Migrants</i>							
Treatment	-0.033 (0.035)	0.011 (0.024)	-0.339 (0.485)	0.023 (0.025)	-1.485 (1.828)	-0.020 (0.037)	-0.081** (0.035)
Q-value	1.000	1.000	1.000	1.000	1.000	1.000	0.192
Mean Control	0.705	0.127	9.323	0.0903	23.79	0.424	0.250
MDE	±0.097	±0.068	±1.360	±0.070	±5.122	±0.104	±0.099
Observations	1243	1243	1242	1243	1243	1243	1243
Competence FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Wave FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Treatment is an indicator equal to one for individuals randomly assigned to receive a skill certification, regardless of whether certification was completed (intent-to-treat, ITT). The table reports pooled ITT estimates from Equation 1, stacking three follow-up rounds at 3, 6, and 12 months. Panel A presents results for the full sample. Panels B and C report estimates separately for locals and migrants. Outcomes in columns (3)–(7) are set to zero when the individual is not employed; monthly earnings are set to zero before applying the log transformation $\log(\text{wage} + 1)$. All specifications include competence and wave fixed effects; full-sample regressions additionally control for migrant status. Standard errors (in parentheses) are clustered at the individual level. Sharpened q -values are reported for Panels A, B and C and are computed within each outcome family following [Anderson \(2008\)](#). MDE reports the minimum detectable effect at the 5% significance level with 80% power, computed using the standard error of the treatment effect. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.2 Dynamic Effects

The pooled estimates may mask differences in timing and persistence. Effects may emerge quickly if access to the certification program improves job matching during search, or more gradually if employers learn about certified workers over time. They may also attenuate if the informational value of certification declines as employers acquire direct information about workers' productivity through labor market

employment, hours worked, or wages. We find suggestive evidence of a reduction in skill downgrading, but this estimate does not survive the multiple-hypothesis adjustment and partly reflects changes in employment composition. We therefore do not interpret the migrant results as establishing an improvement in occupational matching. More generally, the experiment provides limited evidence on how certification affects migrants and lacks the statistical power to rule out economically meaningful effects for this group.

These findings have two implications. First, credible certification of existing skills can affect labor-market outcomes even without providing training. Second, the effects of certification can differ substantially across groups. In our setting, the evidence is strongest for employment gains among locals, while the migrant estimates are less precise and do not show comparable gains in employment, hours worked, or wages. This heterogeneity matters for policy design and evaluation: average effects may mask important differences across target populations, and understanding the sources of those differences requires evidence beyond the treatment effects themselves.

Several limitations remain. The estimates come from one cohort in Colombia's South-western region and cover outcomes up to twelve months after certification. The migrant sample is smaller than the local sample, which limits precision for some outcomes. The experiment also does not directly observe employer beliefs or hiring behavior, so the signaling interpretation remains indirect. Future work that measures employer responses to certified and non-certified applicants would help distinguish more directly between demand-side screening, worker search behavior, and other mechanisms.

Environmental and Gender Impacts of Land Tenure Regularization in Africa

Pilot evidence from Rwanda

Daniel Ayalew Ali

Klaus Deininger

Markus Goldstein

The World Bank
Development Research Group
Agriculture and Rural Development Team
&
Africa Region
Gender Team
August 2011



Abstract

Although increased global demand for land has led to renewed interest in African land tenure, few models to address these issues quickly and at the required scale have been identified or evaluated. The case of Rwanda's nation-wide and relatively low-cost land tenure regularization program is thus of great interest. This paper evaluates the short-term impact (some 2.5 years after completion) of the pilots undertaken to fine-tune the approach using a geographic discontinuity design with spatial fixed effects. Three key findings emerge from the analysis. First, the program improved land access for legally married women (about 76 percent of married couples) and prompted

better recordation of inheritance rights without gender bias. Second, the analysis finds a very large impact on investment and maintenance of soil conservation measures. This effect was particularly pronounced for female headed households, suggesting that this group had suffered from high levels of tenure insecurity, which the program managed to reduce. Third, land market activity declined, allowing rejection of the hypothesis that the program caused a wave of distress sales or widespread landlessness by vulnerable people. Implications for program design and policy are discussed.

This paper is a product of the Agriculture and Rural Development Team, Development Research Group. It is part of a larger effort by the World Bank to provide open access to its research and make a contribution to development policy discussions around the world. Policy Research Working Papers are also posted on the Web at <http://econ.worldbank.org>. The authors may be contacted at dali1@worldbank.org, kdeininger@worldbank.org or mgoldstein@worldbank.org.

The Policy Research Working Paper Series disseminates the findings of work in progress to encourage the exchange of ideas about development issues. An objective of the series is to get the findings out quickly, even if the presentations are less than fully polished. The papers carry the names of the authors and should be cited accordingly. The findings, interpretations, and conclusions expressed in this paper are entirely those of the authors. They do not necessarily represent the views of the International Bank for Reconstruction and Development/World Bank and its affiliated organizations, or those of the Executive Directors of the World Bank or the governments they represent.

Environmental and Gender Impacts of Land Tenure Regularization in Africa:

Pilot evidence from Rwanda

Daniel Ayalew Ali, Klaus Deininger, Markus Goldstein*

JEL classification: J16, O13, Q12, Q15

Keywords: Gender, agricultural investment, land administration, Rwanda

* The World Bank, Ali: dali1@worldbank.org; Deininger: kdeininger@worldbank.org; Goldstein: mgoldstein@worldbank.org. We thank the UK Department for International Development, the Global Land Tools Network, the Bank Netherlands Partnership Program, and the Gender Action Plan for funding support. We are grateful for insightful conversations with Peter Berck, Stein Holden, Gunnar Kohlin, Jeremy Magruder, Loraine Ronchi, Zurab Sajaia, Chris Udry and for comments from seminar and conference participants the University of California Berkeley, the Norwegian University of Life Sciences, the World Bank's 2011 Annual Conference on Land and Poverty, UNU-WIDER, and Rwanda's Agricultural Sector Steering Group as well as the Rwanda Natural Resource Authority's Land Thematic Group. The work would not have been possible without the collaboration and guidance of the National Land Centre in Kigali, especially Emmanuel Nkurunziza and Didier Sagashya. The views presented in this paper are those of the authors and do not represent those of the World Bank or its member countries.

1. Introduction

While Africa has been historically rather land abundant, three factors have recently focused attention on land rights and the way in which land is managed. First, a combination of population growth, soil degradation, urban expansion, exacerbated more recently with a ‘rush’ for land to benefit from rising global demand for agricultural commodities and environmental services, increased pressures on land. Second, even lands currently cultivated are characterized by an enormous productivity gap; recent studies show that no African country except South Africa achieves even 25% of the potential (Deininger *et al.* 2011b). The investments needed to bridge this gap are unlikely to be forthcoming without secure rights. Third, structural transformation from an agrarian to a manufacturing- and service-based economy will require shifts of population and associated transfers of land to other producers. Given their potential to contribute to all of these objectives, land registration programs should be high up on the policy agenda. Yet, limited success and sustainability of many past programs, together with a lack of mechanisms to document impact over time, led many observers to believe that such interventions will have limited benefits or be too complex and risky to undertake.

As the most densely populated country in Africa, Rwanda illustrates the relevance of the above issues. Recognizing the urgency of solving them, the government has taken far-reaching measures to overcome a history of often land-related conflict and tribal division, end gender discrimination in land access, and provide a framework to make optimum use of available land resources so as to contribute to social and economic development. In this context, the country has launched a program of land tenure regularization (LTR) which, after extensive piloting, is now being implemented nation-wide. This program constitutes probably the most thoroughly designed and quantitatively ambitious program of its nature in Africa. An evaluation of the extent to which it lives up to its objective of securing rights and improving opportunities and livelihoods for all Rwandans can thus provide broader lessons.

We aim to provide such an evaluation by assessing the impact of the (rural) pilots that preceded the program’s national roll-out. Outcome variables considered are the incidence of land-related investment, female land ownership and inheritance, and the frequency of land transactions. We find that, in the short time since its implementation, the intervention has led to a doubling of the change in investment in soil conservation, with an even larger increase for females. Formalization of women’s land rights helped to expand their land access and tenure security while clarification of intended inheritance increased clarity. The program is thus likely to allow the country to realize sizeable economic and gender benefits. As the program did not lead to any increase in land market activity, it is unlikely to have contributed to distress sales or increased landlessness.

The program's positive impacts can be enhanced—and potential negative ones avoided—if areas where policy is unclear, ambiguous, or at variance with practice on the ground, are addressed and performance in high risk areas is carefully and continuously monitored. Based on our results these include (i) land rights by women who are not legally married; (ii) fee structures that rural residents cannot afford and subdivision restrictions that the majority of land holders will not be able to comply with; and (iii) systems to reduce the transaction cost of registering subsequent transactions so as to ensure sustainability. All of these areas are being discussed by the government in the context of program implementation and an ongoing review of the land law. Beyond Rwanda, our evidence suggests that well-designed land regularization can yield large benefits even in the short term but that realization of these will require a conducive policy context, careful design based on local conditions, and continued monitoring during implementation.

The paper is organized as follows. Section two places the topic in context and identifies areas that warrant greater empirical attention by reviewing lessons from the literature on land titling in Africa. Section three describes the challenges confronted by Rwanda, the institutional changes made in response to these, and relevant characteristics of the LTR program. Econometric approach and sampling strategy are described in section four, followed by a discussion of key results with respect to investment, women's land rights, inheritance, and land market participation in section five. We conclude with a number of policy implications in section six.

2. Background and context

While issues of land tenure formalization in Africa have long been the subject of debate, increased land scarcity and recent growth in demand for land reinforce the importance of securing rights to allow land-related investment that would close the productivity gap while at the same time ensuring that increased land values benefit local land users rather than raising the specter of dispossession. A review of the pros and cons of land titling interventions as evidenced by the conceptual and empirical literature sets out the challenges which any program of land tenure regularization will have to confront.

2.1 The debate on land tenure formalization in Africa

Given the economic and social importance of land, the desirability, nature, and impact of interventions to increase security of land tenure in Africa have long been intensely debated. Three arguments have traditionally been advanced to caution that such measures may not be needed or may have unintended negative consequences that could outweigh whatever positive impacts they may bring.

First, land tenure systems evolve dynamically over time and many studies have documented that, as payoffs from investment increase, institutional innovations to increase tenure security emerge

impacts of land regularization in Rwanda below will focus on impacts with respect to investment, equity, and land transfers, noting where appropriate the potential impact of obstacles to land market operation.

3. Rwanda's setting

As the most densely populated country in Africa, Rwanda has long faced the challenge of establishing a land system that would provide incentives for investment and rational land use to create economic opportunities while at the same time overcoming gender and ethnic biases in land access. A failure to meet these challenges has been identified as contributing to the 1994 genocide. Since then, enormous effort have been undertaken to clarify land rights and provide the basis for overcoming traditional biases, culminating in the LTR program that is to be evaluated here.

3.1 The challenges posed by land scarcity and conflict

With some 85% of its population deriving their main income from agriculture and the highest population density of any African country (384 inhabitants per km² in total and 526/km² for agricultural land), land-related investment and effective land use are key determinants of poverty reduction in Rwanda (Republic of Rwanda 2009). In 2008, the average household had 0.72 ha of land, slightly below the threshold of 0.75 ha that official sources estimate is needed for a family to satisfy nutritional needs with available technology. Land is highly fragmented; the average household owns four parcels, with average parcel size varying between 0.13 ha in the North and 0.37 ha in the East. Rapid population growth has led farmers to push into marginal lands, clear forests, and cultivate steep hillsides without proper soil and water conservation, creating severe environmental challenges (Republic of Rwanda 2004).

In Rwanda customary land tenure systems had traditionally provided high levels of tenure security. However, outdated processes and misuse, e.g., by state representatives invoking eminent domain even for private benefits, had led to serious deterioration and weakened the system's ability to cope with far-reaching social, economic and political changes. Increased land scarcity and population pressure failed to trigger institutional innovations to encourage greater investment and more intensive land use. Instead competition for land in an environment characterized by slow expansion of non-agricultural income opportunities resulted in illegal land sales, pervasive land disputes, and 'land grabbing' that exacerbated inequality, landlessness, and social tensions. Although they are not the only factors, land-related conflicts have been identified as contributors to the 1994 genocide (Andre and Platteau 1998) and land policy is widely recognized to have a central role in creating preconditions for sustained growth and conflict prevention in today's Rwanda. To rise to this challenge, the country embarked on an integrated strategy of legal and institutional reform, the pilots evaluated here, and eventually nation-wide implementation of the land tenure regularization program.

3.2 Legal and institutional changes

To provide secure land tenure to all Rwandans in the context of national unity and reconciliation, and with the goal of creating the preconditions for rapid structural transformation, far-reaching legal and institutional changes were embarked upon. Adoption of the 1999 inheritance law, key provisions of which were also incorporated into the 2003 constitution, aimed to eliminate bias against female land ownership. It was followed by the 2004 land policy and its codification in the 2005 organic land law (OLL). Establishment of institutional structures at national and local level then created the preconditions for the pilots the evaluation of which is the topic of this paper and a national program.

As is common in customary systems, women in Rwanda formerly had land use rights only through their husbands whose lineage controlled the land, implying that their right to own or inherit land was severely compromised. In fact, widows were unable to inherit their deceased husband's property and at most were allowed to use it until male children grew up. Those without children lost even use rights to family land unless they maintained family ties by marrying one of their husband's brothers (Republic of Rwanda 2004). The 1999 inheritance law changed this by advancing in three areas. First, daughters and sons are granted equal rights to inherit their parents' property. Second, subject to the provisions of the family law (which under the most common conjugal property regime mandates equal shares), property rights by women under a legally registered marriage are protected.² Third, spousal consent is required for transaction (e.g., sale, mortgage or exchange) of matrimonial property by any of the marriage partners. Nevertheless, the law does not protect property rights by women who live in marriages that are unregistered or arrangements, including consensual unions, customary or religious marriages, and polygamous unions, that are not formally recognized.³

After animated debate, the 2004 Land Policy put forward general principles for efficient and sustainable use of scarce land resources and called for a legal and institutional framework to achieve these. The 2005 OLL provides the legal basis for the necessary arrangements. One of its key provisions is to establish a single statutory system of land tenure in order to end the dualism created by parallel existence of customary and formal tenure systems. Ownership of land is vested with the state; landholders are provided long-term, usufruct rights (up to 99 years, depending on land use) that can be sold, passed on to heirs, mortgaged, leased, or otherwise transferred. At the same time, and in line with the Land Policy, further subdivision of agricultural parcels of land less than or equal to one ha is prohibited and any

² Rwanda's Constitution gives legal recognition to registered monogamous marriage between a man and a woman. Any other arrangements such as consensual union, customary or religious marriages are not recognized by law and hence do not get any legal protection and property rights.

³ Anecdotal evidence suggests that, especially for the post-genocide generation, the cost of conducting a 'proper' marriage ceremony is often prohibitive, something that can result in delays during which couples live together informally.

subdivision of parcels between 1 and 5 ha requires approval by a competent authority. The law also lays the basis for land expropriation (with appropriate compensation) in the public interest.

To translate the unification of land tenure systems into reality, the OLL recognizes land acquired through customary law but makes first-time land registration and recording of any follow up transfers compulsory. At national and district levels, land commissions were formed to oversee OLL implementation.⁴ The National Land Center (NLC),⁵ which includes the registrar's office, was established as the technical agency in charge of all activities related to land administration, land use planning and management, and OLL implementation more generally. At district, town, and municipality level, District Land Bureaux (DLBs) are made responsible for land administration and use planning, complemented by land committees at sector and cell levels to serve as focal points for land registration and land use planning and thus facilitate a decentralized and participatory implementation process.

3.3 The national land tenure regularization (LTR) program

To clarify existing land rights on all of the country's estimated 11 million land parcels, the NLC has been tasked with development and implementation of a program of land tenure regularization (Sagashya and English 2010). The goal is to create the precondition for formalization and full legal recognition of rights to these lands, as manifested in the award of title certificates to land holders. Given the lack of successful models that could be drawn upon and the heterogeneity of situations across different parts of the country, it was decided to develop and fine tune the methodology through a pilot exercise in 4 cells, chosen to reflect typical situations encountered throughout the country, before embarking on a national roll-out.

This pilot, which was implemented in 2007/2008, resulted in demarcation and adjudication of 14,908 parcels with a total area of 3,448 hectares, owned by 3,513 households. After declaration of an area as subject to adjudication and conduct of stakeholder sensitization programs, locally trained para-surveyors conducted land demarcation in the field to identify parcel boundaries in the presence of land owners and all adjoining neighbors, mark them on an aerial photo to create a graphical record and, for undisputed parcels,⁶ issue a claim receipt that is signed by all adjoining neighbors. Information from this receipt, in particular the names of all persons, including women and minors, with a claim or interest on the property, is then transferred to a registry book, digitized, and displayed publicly. If no objections are raised within a

⁴ The Republic of Rwanda is composed of five local administrative entities along with that of the central government. Accordingly, the country is divided into four provinces (North, South, East and West) and the city of Kigali, 30 districts (of which three of them are in the city of Kigali), 416 sectors, 2,146 cells, and 14,876 villages (*umudugudu*). The roles and responsibilities of each of the entities are clearly specified in the "Rwanda Strategic Decentralization Framework" (Republic of Rwanda 2007). Provinces are to serve as a liaison "for coordinating district development planning with the national policies and programs" and to supervise their implementation at the district level. The districts play a prominent role in the decentralization process as they are responsible for "local economic development planning and coordinating the delivery of public services". Sectors assume a coordinating role in the delivery of public services and gather data and information. As the smallest administrative unit, cells prioritize needs and mobilize the community to address them. They are composed of villages that do not have specific administrative functions.

⁵ In 2011, the NLC was integrated into the Rwanda Natural Resource Authority as the Department of Land and Mapping.

⁶ Parcels, the ownership claims to which are disputed, are recorded in a separate dispute register. Disputed claims can then be pursued separately through either administrative or judicial channels with the possibility of legal NGOs playing an important role in moving the process along.

period of public display of at least 2 weeks, the information is formally registered, creating the precondition for award of a formal certificate upon payment of a nominal fee.⁷

Figure 1 displays the location of the trial areas on the map, and a brief review of their main characteristics will be of interest (Republic of Rwanda 2008). Nyamugali, located in a densely populated peri-urban area (3,500 inhabitants per km²) in Kigali city, is a largely residential area interspersed with small agricultural plots where a thriving informal land market and high levels of expropriation risk had created demand for legalization of existing land records. Kabushinge cell represents a high population density rural area (768 inhabitants per km²) in the North where high population pressure and land scarcity together with land conflicts arising from high levels of polygamy and post-genocide land allocation to refugees and soldiers created land pressure. Biguhu cell, in a moderately populated Western area (337 persons/km²), represents areas severely affected by the genocide with many female household heads and orphan landholders. Finally, Mwoga cell is in a low population density sector (274/km²) in the East where extensive land sharing and high concentration of refugees caused widespread fear and tenure insecurity. Completion and careful analysis of operational aspects of the pilot experience allowed NLC to scale up to a nation-wide program that currently deploys more than 60 field teams who, in the program's first year, demarcated a total of 4.8 million parcels.

4. Sampling, econometric approach, and data

While the program presents a number of interesting aspects, a key methodological challenge to overcome for a rigorous socio-economic assessment of the pilots was the lack of a baseline data. To deal with this, we adopt a spatial discontinuity design and administered a short survey to some 3,500 households on both sides of the boundaries of the four pilot cells. This section describes the sample design and the econometric methodology, relying on the use of spatial fixed effects that is used to interpret the data.

4.1 Sampling strategy

As there was no usable baseline survey to assess the impact the pilot program, we rely on cross-sectional data, sampled from both sides of the boundaries of the pilot cells to allow comparison of outcome variables between treated and non-treated households (inside and outside the borders of the pilot cells, respectively) close to the borders of the pilot cells. We thus exploit the discontinuity which administrative boundaries created in the introduction of the pilot program as an identification strategy. The key assumption is that before the start of the program, households close to a cell boundary were similar in unobservable and observable factors affecting relevant outcomes.

⁷ The fee of RWF 1000 (US\$ 1.84 at the 2008 rate) per parcel or about RWF 4000 (US\$ 7.36) per household taking the national average of four parcels per household compares to a cost of approximately US\$ 9-11 in the pilot—since then reduced to about US\$ 5—per parcel.

To obtain household information, we administered a light survey during April-May 2010, about 2.5 years after the start of LTR in the pilot cells. The sample was designed to yield numbers of households in each pilot cell equivalent to their share in the total, with a size of 3,554 households with some 6,330 land parcels, intended to be split equally across pilot and their neighbor cells. To sample inside pilot cells, we used parcel index maps created by the program. As no such maps were available for the adjacent (control) cells, we used high resolution satellite imagery to identify dwellings that could then serve as a sample frame. Figure 2 illustrates the procedure graphically. To ensure that the share of households in each pilot cell corresponded to the cell's share in the target population, the widths of the bands from which households were selected were adjusted.⁸ Note also that, as we lack enough statistical power to make strong conclusion for the urban sample, i.e., the city of Kigali, the empirical analysis given below is solely based on data from rural areas.

Table 1 shows the distribution of sampled households in the pilot and their adjacent cells. Information was collected both at household and parcel levels. Household level information includes demographics, housing, assets, participation in the credit market, participation in the registration program and knowledge of the law and the GPS coordinates of the main residential plot. Parcel level questions included land characteristics, investment, inheritance dynamics as well as participation in land sales and rental markets.

4.2 Econometric approach

To examine the effects of LTR, we estimate the following equation:

$$Y_{ph} = \alpha + \beta_1 T_{ph} + \beta_2 X_{ph} + \beta_3 Z_h + \varepsilon_{ph} \quad (1)$$

where Y_{ph} is the outcome of interest for parcel p in household h , T is an indicator for LTR participation, X is a set of parcel characteristics, Z household characteristics, and ε an error term. If land registration were voluntary, we would be concerned about endogeneity of outcomes such as investment and land market activity. However, tenure regularization was compulsory and covered all private land in the cell. In other words, plots were registered up to a cell's administrative boundaries, but not beyond the cell border. The spatial discontinuity generated by this allocation rule thus can be used to allow identification of program effects by comparing individuals who live within a band on either side of the border following Magruder (2011). This approach extends work comparing enterprises on either side of a state border in the US to look at the effects of minimum wage legislation (Card and Krueger 1994) to explore effects of centralized bargaining agreements on employment using geographic and labor force data. For the approach to be valid, communities on either side of the border have to be identical and cell-level effects that could drive our results need to be absent. To assess the validity of this assumption, we note that key policies with

⁸ Outer and inner bandwidths were 125 m in Nyamugali, 400 m in Kabushinge, 350 m in Biguhu, and 1250 m in Mwoga.

respect to land rights and use are set and enacted by higher levels of government; for example land inheritance policy is determined nationally and soil conservation policies are mostly enacted at the district level. Thus, while the cell administration plays an important role in key aspects of implementing the LTR process such as public sensitization, conflict resolution, display, and registration, all of these are part of the treatment of interest and there are no significant cell level interventions that could drive our results.

We might, however, still be concerned about the possibility of local variations in soil quality, market conditions or other factors that could be driving our results. We follow the literature (Conley and Udry 2010, Goldstein and Udry 2008, Magruder 2011) and use spatial fixed effects to control for local level unobservables. Equation (1) now becomes:

$$Y_{ph} - \frac{1}{N_p} \sum_{q \in N_p} Y_{qh} = \beta_1 \left(T_{ph} - \frac{1}{N_p} \sum_{q \in N_p} T_{qh} \right) + \beta_2 \left(X_{ph} - \frac{1}{N_p} \sum_{q \in N_p} X_{qh} \right) + \beta_3 \left(Z_h - \frac{1}{N_p} \sum_{q \in N_p} Z_{qh} \right) + \varepsilon_{ph} - \frac{1}{N_p} \sum_{q \in N_p} \varepsilon_{qh} \quad (2)$$

where N_p is both the set of parcels within a critical distance of parcel p and the number of such parcels. In other words, (2) provides a continuous way to compare treatment and control households whereby each household is combined with those in a neighborhood (some of whom will be literally next door) which are some combination of treatment and control households. As noted above, we interviewed all households in the band close proximity to the cell border (i.e., inside and outside) to make it more likely that households are similar. Indeed, while this approach will give us a local treatment effect only, it is ideally suited for spatial fixed estimations by delivering a high density of treatment and control households in close proximity to one another. As explained above, we do not have geographic data on the location of individual parcels. Instead, we define the critical distance based on a neighborhood of households. To ensure that we capture all of a household's parcels and in order to ensure that each treatment (control) household's neighborhood contains at least one control (treatment) household for comparison, we defined this critical distance as 1000m. Finally, to deal with possible spatial autocorrelations in the error term, we will use the method of Conley (1999) to correct the standard errors.

5. Results

Tables 2 and 3 present summary statistics by treatment status. Table 3 points towards high levels (22%) of female headship, small landholdings (0.93 ha for the average household), and formalized marriages by about three quarters (76.4%) of the sample. It also documents that for most of the time-invariant household characteristics of interest, treatment and control households are not different from one another in any meaningful way. However, for some of the characteristics that might be affected by the program,

such as mode of acquisition of a parcel and length of possession (Table 3), significant differences emerge. Since this is cross-sectional data, it is quite possible that some of these differences may be driven by the program. For example, the fact that average length of possession is higher in the treatment area than the control is consistent with the result below that land markets are less active in the treatment area.

Table 4 displays the mean values of key outcome variables of interest. Contrary to the earlier tables, there are highly significant differences between program and control areas that may point towards program impacts. However, use of spatial fixed estimates (and correction of standard errors for spatial dependence) will be required to draw robust conclusions regarding the program's impact. As there are indeed a number of cases where differences suggested by descriptive tables lose significance once a proper methodological framework is applied, we discuss these variables in the context of the econometric estimations below.

5.1 Land related investment

A primary reason for the Government of Rwanda to initiate LTR was to increase levels of tenure security. Table 5 reports evidence on program impacts on subjective expropriation risk using a binary measure and the five-scale grade included in the questionnaire.⁹ While the sign on the treatment coefficient is negative as expected, it is not statistically significant. Possible reasons include the fact that LTR does not change overall policy and that households' knowledge of the policy may be limited, something that could be addressed in a follow-up by asking more specific questions on knowledge of expropriation procedures.

In contrast to the above result, table 6 documents household behavior that is consistent with a lengthening of time horizons due to LTR-induced increases in tenure security. Consistent with the notion that improved seeds do not yield long-term benefits, the coefficient of this variable in column 1 is positive but not statistically significant. At the same time, column 2 suggests that households affected by LTR are almost 10 percentage points more likely to make or maintain soil conservation investments in structures such as bunds, terraces, and check dams. This is about double the change in investment in the control group, a very large effect. Women seem to benefit more in this respect; estimated effects of LTR on such investment by female headed households is double that of men with female headed households exhibiting a roughly 19 percentage point increase in likelihood of measures to construct or maintain soil conservation structures. This suggests that low levels of tenure security by females acted as an obstacle to investment by this group and that removing such impediments by increasing women's tenure security and formalizing rights which they may have enjoyed on an informal basis.

⁹ The wording of the question was "What is the likelihood that you will lose this parcel due to expropriation in the coming 5 years". The binary index is set to one for anybody indicating some expropriation risk.

5.2 Women's land rights and inheritance

A more direct test of the extent to which Rwanda's policy of actively increasing women's rights to land had the desired impact is to assess the extent to which LTR helped to increase female land ownership. However, a key provision of the legislation is that legally married women are entitled to equal ownership rights over household parcels. 76 percent of couples in our sample have a marriage certificate, in line with legal provisions requiring registration of marriages at the local commune. Table 7, which presents regressions for an indicator of land ownership and the share of land owned by females, illustrates that LTR has dramatically changed women's rights over land by helping to create documentary evidence of such rights. While the first four columns, based on the full sample, show little change in female land ownership rights, columns 5 and 6 which restrict the sample to cohabitating/married couples, reveal a large program impact. The first row shows that, for women in this group who are not legally married, LTR results in a small but statistically significant reduction (by 8 percentage points) of the probability of having documented land ownership.¹⁰ However, for women who are part of a union formalized through a marriage certificate, the effect of the program is overwhelmingly positive—they are 17 percentage points more likely to be regarded as joint land owners after LTR than before. The final column displays results with the share of the land owned by women as the dependent variable which point to a positive but not statistically significant effect, regardless of the presence of a certificate.

As many past land adjudication efforts failed to tackle inheritance issues, it is of interest to explore LTR-induced effects in this regard. Table 8 highlights that, most likely because it requires an explicit record of who will inherit the parcel, the process significantly reduces succession-related uncertainty. We find a 9 percentage point increase in the respondent's response that she/he now knows who will inherit the parcel. Column 2 shows that greater certainty along these lines strongly benefits respondents' the children and that gender bias has virtually been eliminated, at least for couples. In LTR areas, children are 13 points more likely to inherit the land and, as columns 4 and 5 document, girls both benefit almost equally. Column 4 suggests that girls in female headed households are significantly less likely to be designated as an heir compared to their counterparts in male headed households. This result deserves further research, but it is not inconsistent with data from Mexico which shows that women exhibit strong and persistent preferences for male versus female heirs (Deere and Leon 2003).

5.3 Land market activity

It is often claimed that, by reducing in the transaction costs of transferring land, interventions to improve land tenure may set off a wave of distress sales whereby households who lack access to credit and

¹⁰ While our questionnaire did not elicit information on disputed claims, it would in principle be straightforward to use the dispute register to explore how many of these were able to register disputed claims and how many of these had actually been resolved.

insurance sell off their land in periods of distress well below its fair market value, thereby entering into a downward spiral of impoverishment. Results from testing this directly with our data as reported in table 9 allow us to convincingly reject this hypothesis for our sample; instead of an increase, we find a statistically significant reduction in land market activity and the size of land area traded due to LTR.

To interpret this finding, two contextual factors are relevant. First, in early 2008, i.e., during pilot implantation, stamp duty, which is the fee to be paid upon registration of a transfer, was changed from 6% of the property value to a flat fee of RwF 20,000 which, for smaller plots, could easily exceed 25% of the land value.¹¹ Second, at the time of the survey, a formal system to register land transfer was not yet in place and it was unclear to what extent the prohibition on registering transfers of parcels below the legally prescribed minimum size of 1 ha—which would have affected virtually all the transfers in our sample—would be enforced. Thus one possibility for the reduced level of land market activity is that households refrained from land transfers in an attempt to not jeopardize their newly acquired tenure security, especially if sensitization in the course of program implementation had increased their awareness of registration fees. Another possibility is that, since this was a pilot that preceded the national roll-out by approximately 2.5 years, the benefits from registration in land markets were still unclear and beneficiaries preferred holding on to their land in expectation of increased land values as others would realize the benefits from having secure land ownership documentation. This is an area that will be more fully explored in upcoming work on the national roll-out of this program.

6. Conclusion and policy implications

Taken together, our results point towards significant benefits from land tenure regularization in Rwanda. The positive results are even more impressive in light of the fact that the program analyzed here was a pilot that involved considerable learning and that the period elapsed between its completion—in particular the actual award of titles—and our survey had been quite short. Individuals whose parcels had been registered through LTR, in particular female-headed ones, were much more likely to invest in soil conservation measures on their land. Clarification and documentation of rights reduced uncertainty over who would inherit land, with substantial benefits for female children who might otherwise have been discriminated against. This program also provided large additional gender benefits. Legally married women were significantly more likely to have their informal ownership rights documented and secured after registration. But women who were not legally married saw diminished property rights, in accordance with the law. And girls residing in female headed households were less likely to be designated as heirs.

¹¹ At the same time, transfers due to inheritance, which had earlier been exempted, were required to pay the same level of fees. In light of the high levels of land market transactions observed in Rwanda, such a fee structure could quickly result in a reversion to informality.

The nature of the survey instrument and the short time since the pilots preclude a full assessment of the productivity and welfare impacts of increased investment, stronger rights by women, and changes in the functioning of land markets. All of these, as well as the program's impact on households' vulnerability and ability to deal with risk, its interaction with other interventions (in particular government programs), and the extent to which modalities of program implementation affect observed outcomes are left for future research. These limitations notwithstanding, our results suggest that LTR indeed focuses on a key constraint to environmental protection, agricultural development, and female empowerment in Rwanda. They also help to identify risks in the areas of policy, implementation processes, and sustainability which, if not attended to in a proactive manner, might jeopardize or at least attenuate the program's positive impact or undermine its sustainability. Follow-up research will be of great interest to document how these risks are addressed, explore in greater detail the program's social, environmental, and productivity dimensions, and help to better appreciate the magnitude of long-term program benefits and the channels through which they materialize.

Table 5: Spatial Fixed Effects Estimates of the Impact of LTR on Perceived Risk of Expropriation

	Binary measure	Five scale measure
Treatment indicator	-0.047 (0.997)	-0.204 (1.251)
Treatment X Female headship	-0.042 (0.512)	-0.106 (0.478)
Number of years possessed	-0.000 (0.411)	-0.001 (0.454)
Parcel was purchased	-0.053* (1.911)	-0.219** (2.333)
Parcel was inherited	-0.073* (1.905)	-0.317** (2.447)
Acquired through other means	-0.004 (0.056)	-0.273 (1.132)
Parcel size in hectares	-0.010 (1.254)	-0.044** (2.128)
Head's age	0.000 (0.295)	0.001 (0.246)
Female headed household	-0.066 (1.444)	-0.231* (1.653)
Head has at least primary education	0.005 (0.335)	0.051 (0.922)
Number of male children age 14 and less	-0.015* (1.661)	-0.056** (2.108)
No. of female children age 14 and less	-0.019* (1.775)	-0.052 (1.643)
Number of adult male age 15 to 60	0.010 (0.828)	0.001 (0.032)
Number of adult female age 15 to 60	0.019* (1.830)	0.044 (1.383)
Male members age 60 and above	-0.022 (0.434)	-0.075 (0.495)
Female members age 60 and above	-0.074 (1.451)	-0.226 (1.347)
Constant	-0.001 (0.165)	-0.015 (0.497)
Number of observations	5345	5345
R2	0.030	0.032

Note: The geographical neighborhood of each household is defined at a distance of 1000 meters. Absolute value of t-statistics corrected for spatial dependence (1500 meters). * significant at 10%; ** significant at 5%; *** significant at 1%.

Table 6: Spatial Fixed Effects Estimates of the Impact of LTR on Rural Investment

	Change in improved seed use	Change in soil conservation measures		
		New or maintenance	New construction	Maintenance
Treatment indicator	0.064 (1.008)	0.099** (2.460)	0.075 (1.544)	0.038 (1.271)
Treatment X Female headship	0.003 (0.076)	0.094** (2.076)	0.026 (0.713)	0.050 (1.314)
Number of years possessed	0.001 (1.187)	-0.002** (1.988)	-0.000 (0.194)	-0.002*** (2.645)
Parcel was purchased	-0.074** (2.438)	-0.030 (0.901)	-0.018 (0.623)	-0.017 (0.940)
Parcel was inherited	-0.083 (1.373)	-0.065 (1.030)	-0.016 (0.334)	-0.052* (1.705)
Acquired through other means	0.087 (1.226)	-0.209** (2.512)	-0.133 (1.640)	-0.081* (1.732)
Parcel size in hectares	0.032* (1.826)	-0.002 (0.199)	-0.001 (0.088)	-0.004 (0.650)
Head's age	-0.001 (0.927)	-0.001 (0.978)	-0.001 (0.720)	-0.001 (0.762)
Female headed household	-0.052* (1.668)	-0.044 (1.183)	-0.040 (1.144)	-0.003 (0.094)
Head has at least primary education	0.005 (0.268)	-0.002 (0.118)	0.005 (0.311)	-0.014 (0.805)
Number of male children age 14 and less	0.002 (0.176)	0.019** (2.017)	0.011 (1.160)	0.018** (2.366)
No. of female children age 14 and less	-0.005 (0.430)	-0.003 (0.227)	-0.004 (0.519)	-0.007 (0.829)
Number of adult male age 15 to 60	0.020* (1.936)	0.020* (1.674)	0.009 (0.868)	0.016* (1.802)
Number of adult female age 15 to 60	-0.002 (0.146)	0.004 (0.278)	0.004 (0.335)	0.009 (0.829)
Male members age 60 and above	-0.041 (0.910)	0.001 (0.017)	-0.013 (0.301)	0.009 (0.256)
Female members age 60 and above	0.046 (1.095)	0.070* (1.872)	0.060* (1.832)	0.037 (1.094)
Constant	0.008 (0.619)	0.003 (0.289)	0.003 (0.308)	0.001 (0.174)
Number of observations	6325	6325	6325	6325
R2	0.014	0.020	0.010	0.016

Note: The geographical neighborhood of each household is defined at a distance of 1000 meters. Absolute value of t-statistics corrected for spatial dependence (1500 meters). * significant at 10%; ** significant at 5%; *** significant at 1%.

Table 7: Spatial Fixed Effects Estimates of the Impact of LTR on Female Access to Land

	Full Sample				Sample of Married Couples	
	Owns land	Share of land	Owns land	Share of land	Owns land	Share of land
Treatment indicator	-0.030 (-1.644)	2.706 (1.127)	-0.033 (-1.633)	1.476 (0.788)	-0.080** (-2.308)	0.438 (0.263)
Treatment X Female headship			0.016 (0.600)	6.309 (1.276)		
Treatment X Marriage certificate					0.170*** (3.057)	4.355 (1.269)
Has marriage certificate					0.076*** (2.775)	3.179** (1.974)
Number of years possessed	-0.002** (-2.059)	-0.030 (-0.726)	-0.002** (-2.061)	-0.030 (-0.731)	-0.001* (-1.873)	-0.079** (-2.151)
Parcel was purchased	0.003 (0.227)	-1.724 (-1.535)	0.003 (0.203)	-1.829* (-1.673)	0.012 (0.998)	-1.646 (-1.545)
Parcel was inherited	0.017 (1.063)	-2.170 (-1.333)	0.017 (1.044)	-2.306 (-1.458)	0.019 (1.372)	-2.335 (-1.622)
Acquired through other means	0.013 (0.436)	8.148*** (3.588)	0.013 (0.444)	8.241*** (3.539)	0.044** (2.195)	7.500*** (3.598)
Parcel size in hectares	0.001 (0.097)	-0.085 (-0.162)	0.001 (0.104)	-0.065 (-0.125)	0.005 (0.693)	0.025 (0.086)
Head's age	-0.001 (-1.070)	0.103 (1.430)	-0.001 (-1.056)	0.106 (1.483)	0.002* (1.663)	0.246*** (2.762)
Female headed household	0.096*** (5.281)	29.886*** (11.119)	0.089*** (4.193)	27.317*** (12.064)		
Head has at least primary education	0.006 (0.388)	-0.035 (-0.035)	0.006 (0.390)	-0.017 (-0.017)	0.009 (0.628)	-0.633 (-0.669)
Number of male children age 14 and less	0.047*** (6.250)	1.293*** (2.930)	0.047*** (6.288)	1.329*** (3.028)	0.010 (1.499)	-0.490 (-1.285)
No. of female children age 14 and less	0.046*** (5.926)	1.137** (2.389)	0.046*** (5.915)	1.187** (2.445)	0.007 (1.060)	-0.742 (-1.566)
Number of adult male age 15 to 60	0.020** (2.112)	-1.294 (-1.500)	0.020** (2.133)	-1.265 (-1.504)	-0.002 (-0.175)	-1.370** (-2.018)
Number of adult female age 15 to 60	0.082*** (7.496)	2.190*** (3.429)	0.082*** (7.511)	2.140*** (3.474)	0.027* (1.884)	0.807 (0.871)
Male members age 60 and above	-0.071** (-1.984)	-11.030*** (-4.068)	-0.072** (-1.973)	-11.152*** (-4.104)	-0.101*** (-2.765)	-9.894*** (-3.193)
Female members age 60 and above	0.317*** (8.623)	16.317*** (8.604)	0.317*** (8.651)	16.593*** (9.002)	0.129*** (2.708)	8.058*** (2.675)
Constant	-0.001 (-0.248)	0.181 (0.392)	-0.001 (-0.245)	0.188 (0.403)	0.023* (1.939)	-3.495*** (-5.960)
Number of observations	6,225	6,219	6,225	6,219	4,609	4,608
R2	0.149	0.348	0.149	0.350	0.120	0.100

Note: The geographical neighborhood of each household is defined at a distance of 1000 meters. Absolute value of t-statistics corrected for spatial dependence (1500 meters). * significant at 10%; ** significant at 5%; *** significant at 1%.

Table 8: Spatial Fixed Effects Estimates of the Impact of LTR on Inheritance Dynamics

	Know who will inherit	Children inherits	Son inherits	Daughter inherits
Treatment indicator	0.094** (2.186)	0.133** (2.276)	0.102** (2.552)	0.096** (2.039)
Treatment X Female headship	-0.044 (0.741)	-0.046 (0.779)	-0.052 (0.735)	-0.158** (2.326)
Number of years possessed	0.001 (0.745)	0.000 (0.112)	0.001 (0.708)	0.001 (0.396)
Parcel was purchased	-0.008 (0.344)	-0.021 (0.964)	0.043 (1.642)	0.008 (0.361)
Parcel was inherited	0.026 (0.719)	0.004 (0.136)	0.072** (2.247)	0.038 (0.995)
Acquired through other means	-0.051 (0.369)	0.025 (0.276)	0.167* (1.928)	-0.030 (0.254)
Parcel size in hectares	-0.023** (2.375)	-0.012 (1.248)	0.006 (0.567)	0.004 (0.284)
Head's age	0.001 (0.820)	0.003** (2.347)	0.002 (1.285)	0.004** (2.476)
Female headed household	0.071* (1.789)	0.094*** (3.003)	0.210*** (5.764)	-0.003 (0.062)
Head has at least primary education	0.018 (0.818)	0.024 (1.337)	0.031 (1.297)	-0.001 (0.025)
Number of male children age 14 and less	0.028*** (2.779)	0.053*** (5.292)	0.155*** (11.481)	-0.017 (1.417)
No. of female children age 14 and less	-0.003 (0.303)	0.028*** (3.178)	-0.041*** (3.064)	0.161*** (13.739)
Number of adult male age 15 to 60	0.014 (1.319)	0.029** (2.362)	0.144*** (10.160)	-0.056*** (3.165)
Number of adult female age 15 to 60	0.010 (0.729)	0.014 (1.157)	-0.040** (2.499)	0.137*** (9.142)
Male members age 60 and above	0.068 (1.296)	0.078** (1.992)	0.123*** (3.077)	-0.160** (2.113)
Female members age 60 and above	-0.035 (0.600)	-0.027 (0.481)	0.026 (0.385)	0.091 (1.572)
Constant	-0.002 (0.209)	-0.002 (0.237)	0.228*** (10.160)	0.212*** (11.360)
Number of observations	6325	6325	4053	4053
R2	0.018	0.057	0.345	0.057

Note: The geographical neighborhood of each household is defined at a distance of 1000 meters. Absolute value of t-statistics corrected for spatial dependence (1500 meters). * significant at 10%; ** significant at 5%; *** significant at 1%.

Table 9: Spatial Fixed Effects Estimates of the Impact of LTR on Land Market Participation

	Change in participation	Change in traded area
Treatment indicator	-0.048** (1.970)	-0.054*** (2.787)
Treatment X Female headship	-0.020 (0.456)	0.044 (1.498)
Head's age	0.001 (1.563)	0.001 (1.111)
Female headed household	-0.002 (0.044)	-0.003 (0.153)
Head has at least primary education	0.021 (0.963)	-0.002 (0.064)
Number of male children age 14 and less	-0.001 (0.116)	-0.004 (0.633)
No. of female children age 14 and less	0.003 (0.324)	-0.002 (0.286)
Number of adult male age 15 to 60	0.002 (0.187)	0.008 (0.691)
Number of adult female age 15 to 60	-0.015 (1.496)	-0.007 (0.714)
Male members age 60 and above	-0.027 (0.811)	0.003 (0.126)
Female members age 60 and above	-0.009 (0.473)	-0.041 (1.046)
Constant	0.000 (0.001)	0.001 (0.189)
Number of observations	2267	2258
R2	0.003	0.004

Note: The geographical neighborhood of each household is defined at a distance of 1000 meters. Absolute value of t-statistics corrected for spatial dependence (1500 meters). * significant at 10%; ** significant at 5%; *** significant at 1%.